Conditional Generation with Classically Trainable IQP Circuits

Oriol Balló-Gimbernat[1,2,3] and Emiliano Tolotti[4]

[1]Eurecat, Centre Tecnològic de Catalunya, Barcelona, Spain
[2]Centre de Visió per Computador (CVC), Barcelona, Spain
[3]Universitat Autònoma de Barcelona (UAB), Barcelona, Spain
[4]Department of Information Engineering and Computer Science, University of Trento, Via Sommarive 9, 38123 Trento, Italy

Quantum circuit Born machines offer expressive generative models, but it is unclear how to make them conditional while preserving the structure that makes some instantaneous quantum polynomial-time (IQP) circuits classically trainable. We preserve that structure with a simple idea: encode the condition in a product-state input, while using the same compiled IQP circuit and the same learned phases for every condition. A single circuit then represents an entire conditional distribution, and its kernel maximum mean discrepancy objective and gradients remain expressible through classically estimable Pauli observables. This extends classically trainable IQP Born machines to conditional generation without requiring classical simulation of the full output distribution during optimization. We study the model on a controlled generative task in which the peak of a Gaussian distribution over Hamming weight moves continuously with the condition. A second benchmark asks whether the same mechanism follows the temperature response of the Curie–Weiss model, whose magnetization law changes from a central profile to an ordered bimodal profile. Across 8–19 qubits and several target widths, the conditioned circuit moves the generated Hamming-weight profile with the requested peak and consistently outperforms an otherwise identical model that receives no condition. This behavior extends to peak locations not seen during training, demonstrating within-range interpolation. Separate circuits trained for individual peak locations are usually more accurate, quantifying the fidelity cost of using one shared circuit. These results establish product-state conditioning as a practical route from classically trainable IQP generation to conditional generation on this benchmark. On Curie–Weiss targets, conditioning improves over the no-condition control at 12 and 16 spins but not at 8 spins. Independently trained neighboring models remain more accurate at every size, and an unrestricted phase model outperforms the symmetry-tied model. Thus the same mechanism produces a temperature-dependent response, but does not by itself deliver high-fidelity thermal generation near the unseen critical point or the cold endpoint.

Oriol Balló-Gimbernat: oriol.ballo@eurecat.org
Emiliano Tolotti: emiliano.tolotti@unitn.it

1 Introduction

Conditional generative models represent a family $q_\theta(x \mid c)$ rather than a single probability law. Quantum circuit Born machines (QCBMs) encode discrete distributions directly in the measurement statistics of parameterized quantum states and have been studied with likelihood-free, kernel, and divergence-based objectives [1–4]. Conditional QCBMs have also encoded external variables by data-reuploading rotations or additional condition qubits [5, 6]. Those works establish that quantum generators can be conditioned; the unresolved question here is whether conditioning can be added to the recently developed *classically trained* IQP Born machine without changing its compiled sampling circuit or its trained phases from one condition to another.

The distinction between training and sampling is central to this question. For commuting circuits, selected observables can be estimated classically under structural assumptions even though

approximate sampling from other IQP families is linked to complexity-theoretic hardness [7–12]. Recent IQP-QCBM work turns that separation into a concrete "train classically, deploy quantumly" program: an MMD objective written in Pauli-Z characters is optimized on classical hardware before the learned circuit is sampled on a quantum device [13, 14]. Our use of *classically trainable* refers only to this observable-estimation interface; it does not assert efficient classical sampling from the learned distribution.

Our construction encodes a scalar condition by broadcast R_Y rotations, then applies one complete two-local IQP circuit. The product input preserves the tractable reference distribution required by the Pauli estimator, while the Hamming radial-basis kernel admits the same character decomposition used for IQP-QCBM training. Input-dependent rotations are established condition- encoding primitives in quantum circuit learning and feature-space models [15–17]. Circuit-Fourier theory describes the resulting condition response as a finite Fourier series [18–20]; the IQP active-generator expansion further filters which frequencies enter a selected observable [14]. We use these established results to explain the model, rather than presenting a new Fourier theorem.

Our contribution is therefore a conditional extension of the classically trained IQP-QCBM framework. The condition changes only the prepared product state, so one trained and compiled circuit represents an entire family $q_\theta(x \mid c)$. We first make that fixed-circuit requirement precise and show how the existing estimator and response descriptions carry over. We then test the construction on a controlled family of Gaussian Hamming-weight distributions. Across sizes and widths, the shared circuit reliably tracks a moving target and interpolates to unseen intermediate conditions relative to a matched circuit that receives no condition. Separate circuits trained for individual neighboring conditions remain a stronger fidelity reference, revealing the cost of sharing one circuit rather than obscuring it. We then challenge the same conditioning mechanism with Curie–Weiss Gibbs laws. This family changes shape with inverse temperature, so it tests qualitative conditional response rather than another translated peak. We test exchange-symmetry parameter sharing as a physical inductive bias without reducing the compiled complete IQP gate sequence. The experiment also exposes its cost: enforcing the correct symmetry need not justify tying the phases so aggressively.

We use *conditional generalization* once in a deliberately narrow sense: interpolation to conditions unseen during training but strictly inside the training interval. We use *interpolation* hereafter. Extrapolation, broader distribution shift, universal conditional expressivity, and quantum advantage are outside scope.

2 Related work and scope of classical trainability

Quantum circuit Born machines. QCBMs are implicit generative models: a parameterized circuit defines a bitstring distribution by Born's rule and produces samples by measurement. Early work developed differentiable MMD training, hardware-oriented shallow circuit learning, and many-body constructions tied to sampling hardness [1–3]. Subsequent studies examined alternative divergences, the quantum-versus-classical learnability of discrete distributions, and generalization beyond memorizing observed bitstrings [4, 21–23]. This paper uses the same Born-machine semantics, but its question is not whether an arbitrary parameterized circuit can be trained from quantum samples. It asks whether one restricted IQP circuit can represent a condition-indexed family while retaining a classical training signal.

Classically trained IQP Born machines. Diagonal commuting circuits occupy a boundary between restricted classical simulation and sampling-hardness results [9, 12, 24]. The IQPopt and train-classical/deploy- quantum frameworks exploit probabilistic estimation of selected observables to optimize large IQP-QCBMs with MMD on classical hardware [8, 13, 14]. Recent analyses sharpen the scope of this promise. Fixed kernels and visible-only IQP models impose expressivity and trainability limits; kernel adaptation and ancillary qubits can enlarge the model class; and gradient scaling depends on topology, kernel spectrum, and initialization [25–28]. Spectral Born machines generalize the same classical-training principle through group Fourier structure [29]. We do not alter those optimization or sampling-

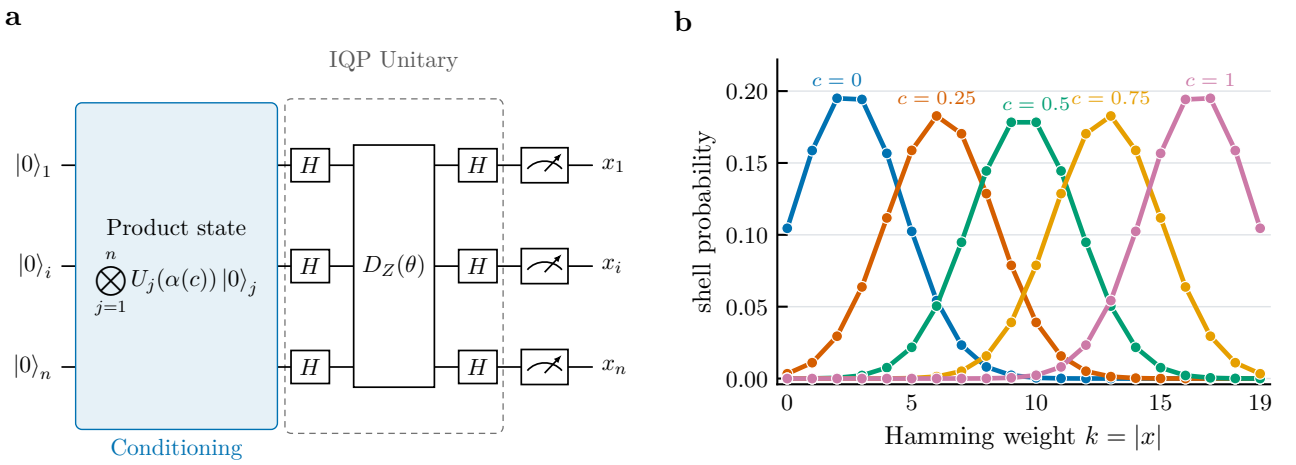


Figure 1: Product-state conditioning and the Gaussian target family. (a) Conditioning prepares the product input $\bigotimes_i U_i(\alpha(c))\,|0\rangle_i$; the IQP unitary and phase vector are shared across conditions. (b) Five representative 19-qubit target profiles; probability is uniform within each Hamming shell.

complexity results. We extend their product-input interface to carry a condition and empirically characterize the resulting shared model.

Conditional quantum generation. Conditional Born machines have generated high-energy-physics events by reuploading an external parameter throughout a variational circuit [5]; that study also evaluates an unseen intermediate condition, so conditional interpolation itself is not our novelty. A separate hybrid construction encodes class labels in additional qubits and combines the circuit with a classical network [6]. More broadly, quantum generative protocols can upload classical randomness or latent variables into parameterized circuits [30]. Our setting differs in the deployment contract: after training, the ordered generator set, compiled gate sequence, and phase vector are identical for every condition; only a separable input preparation changes. This restriction makes the condition a data-loading interface rather than a condition-dependent program.

Encoding, kernels, and terminology. Parameterized quantum circuits embed classical inputs in quantum feature spaces [16, 17]; repeated injection, or data re-uploading, is an established way to enrich their input response [31]. Their response is a Fourier series whose frequencies are controlled by the encoding generators [18–20]. We use that language to describe *response across conditions*, not to claim universal function approximation. The training loss is an MMD two-sample discrepancy [32]. This is a classical positive-definite kernel on bitstrings, not a quantum feature-map kernel; its Hamming-character expansion is what connects the loss to estimable Pauli observables. Throughout the paper, *classically trainable* therefore means that the declared loss and phase gradients are classically estimable at finite budgets. It does not mean that the full bitstring law is efficiently available classically; full-probability simulation and classical output sampling remain outside the claim. Nor does the estimator guarantee an optimization landscape free of barren plateaus [33–36]. Likewise, our use of *interpolation* means evaluation at unseen condition values inside the training interval, whereas QCBM sample generalization concerns unseen bitstrings drawn from a fixed target law [23].

3 Product-state-conditioned IQP model

Let $g_j \in \{0,1\}^n$ identify a diagonal Pauli string $Z_{g_j} = \bigotimes_i Z_i^{g_{j,i}}$. For a fixed ordered generator bank $G = (g_1, \ldots, g_m)$ and phases $\theta \in \mathbb{R}^m$, define

$$D_Z(\theta) = \prod_{j=1}^{m} \exp(i\theta_j Z_{g_j}). \tag{1}$$

For a condition c, the output law is

$$q_\theta(x \mid c) = \left| \langle x | H^{\otimes n} D_Z(\theta) H^{\otimes n} \times \bigotimes_{i=1}^{n} R_Y(\alpha_i(c)) |0\rangle \right|^2 . \quad (2)$$

A model is *fixed-circuit conditional* when the compiled sequence, ordered G, and θ are identical for every condition; only the initial angles may vary. In the main campaign,

$$\alpha_i(c) = \pi(0.35 - 0.70c), \qquad i = 1, \ldots, n. \quad (3)$$

The ansatz contains all one- and two-qubit diagonal generators, hence $m = n(n+1)/2$ phases. The implementation hashes the generator matrix and checkpoint and verifies their invariance across all evaluated conditions.

Two controls clarify what is gained by conditioning. The parameter-matched *no-condition* model receives the neutral input $c = 1/2$ for every target. The *per-condition reference* trains a separate circuit at each training condition. At a held-out coordinate, we compare against (i) a target-aware oracle that selects the more accurate of the two neighboring reference outputs and (ii) a target-blind barycentric mixture of those outputs. If $c_L < c < c_R$, the mixture is

$$q_{\mathrm{mix}}(x \mid c) = \frac{c_R - c}{c_R - c_L} q_L(x) + \frac{c - c_L}{c_R - c_L} q_R(x). \quad (4)$$

Its weights use coordinates only; target metrics never enter. The adjacent-reference oracle is intentionally generous because it may inspect the held-out target to choose between q_L and q_R.

4 Classical training signals

For the Hamming radial basis kernel $k_\sigma(x, y) = \exp[-d_H(x, y)/(2\sigma^2)]$, draw a mask $a \in \{0, 1\}^n$ with independent coordinates

$$a_i \sim \mathrm{Bernoulli}(q_\sigma), \qquad q_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}. \quad (5)$$

Then

$$k_\sigma(x, y) = \mathbb{E}_a[(-1)^{a \cdot x}(-1)^{a \cdot y}]. \quad (6)$$

Thus squared MMD is an average of Pauli-character discrepancies. Independent model and reference draws yield the standard unbiased U-statistic [32]; differentiation of its finite sampled form gives phase gradients. We average bandwidths $(1, 2, 4)$ and conditions. Operator masks, model/reference samples, target samples, minibatches, and validation streams have separately derived seeds.

This establishes access to the declared training signal, not to every output probability. The production optimizer uses 64 masks and 8,192 model/reference samples per bandwidth in training, and 256 masks and 16,384 samples in validation. Exact probabilities are enumerated only after checkpoint selection for evaluation. At $n = 19$ this controlled audit is feasible, but it is not an efficient asymptotic full-distribution procedure.

5 Established response structure

The following propositions collect established structure in the notation of Eq. (2); derivations appear in the supplement.

Proposition 1 (Affine product-input support)**.** *For fixed G, θ, b, W and output x, if $\alpha(c) = b + Wc$, the support of $c \mapsto q_\theta(x \mid c)$ is contained in*

$$\Omega_{\mathrm{full}}(W) = \left\{ \sum_{i=1}^{n} s_i w_i : s_i \in \{-1, 0, 1\} \right\}, \quad (7)$$

where w_i^{T} is row i of W.

This is a direct specialization of multivariate circuit-Fourier theory [18–20]. The inclusion can be strict because frequencies coincide or coefficients cancel, and it gives no efficient procedure for evaluating the coefficients.

Proposition 2 (Observable filter)**.** *For $F_a(c) = \langle Z_a \rangle_c$, only generators satisfying $a \cdot g_j = 1 \pmod 2$ enter the IQP active-generator expansion. Group their subsets by XOR mask h. A term survives on the real R_Y product input only if $h \wedge a = 0$, and its condition frequencies are signed sums of rows w_i with $i \in \mathrm{supp}(a) \cup \mathrm{supp}(h)$.*

This combines the established active-generator formula [14] with $\langle Y_i \rangle = 0$ on a real product state. Selected observable responses can therefore occupy a smaller space than full probabilities. At fixed encoding width both spaces are finite dimensional, which precludes a claim of universal density in arbitrary continuous conditional laws.

6 Gaussian benchmark and frozen protocol

The target is exchangeable within each Hamming shell. At system size n and condition $c \in [0, 1]$, the requested mean is

$$\mu_n(c) = n(0.15 + 0.70c), \tag{8}$$

and the nominal width is $s\sqrt{n}$ for $s \in \{0.35, 0.50, 0.75\}$. We discretize a Gaussian over weights $0, \ldots, n$ and recalibrate its untruncated location numerically so the truncated distribution has mean $\mu_n(c)$. Its shell probability is then distributed uniformly over the $\binom{n}{k}$ strings at weight k. Profile TVD, the TVD between shell-mass vectors, is the primary metric; raw bitstring TVD, mean absolute error, width absolute error, and exact Conditional MMD are secondary diagnostics.

The factorial crosses $n \in \{8, 12, 16, 19\}$, three widths, and ten matched optimizer/data seeds. Shared conditioned and no-condition models train on $c \in \{0, 1/4, 1/2, 3/4, 1\}$; five separately trained circuits form the per-condition reference. All shared models are evaluated at those five coordinates and the common unseen grid $c \in \{1/16, 3/16, \ldots, 15/16\}$. No held-out samples are generated before exact post-selection evaluation.

Every model receives 2,000 Adam updates with cosine learning-rate decay $0.01 \to 0.001$. The total target minibatch contains 450 strings, balanced across conditions; an independent model also uses 450 samples per update. Validation occurs every 25 updates, and checkpoint selection uses only minimum validation Conditional MMD. Shared and no-condition models have matched initialization and estimator streams. Reference-model streams are tagged by training condition. The factorial contributes 840 trained models.

The condition-count study fixes $n = 19$ and $s = 0.50$. Nested grids with $K \in \{2, 3, 5, 9\}$ are evaluated under 10,000/2,000 training/validation draws per condition and under fixed totals of 18,000/3,600 divided among conditions. Each target has one deterministic 10,000/2,000 master pool; fixed-total datasets are prefixes. Nine separately trained per-condition circuits supply a deliberately generous capacity reference, with nested checkpoints reused only by declared identity. Together with the factorial, the study contains exactly 1,020 trained models.

A 100-update $n = 19, K = 9$ resource preflight recorded 22.02 seconds and 1.96 GB peak resident memory on the execution host, projecting 440.30 seconds per model. This passed the preregistered one-hour and 28 GiB resource limits. These are resource checks, not performance thresholds. Full manifest, dataset, stream, checkpoint, probability, environment, runtime, source, and revision digests are archived.

7 Conditional control and interpolation

We average conditions within each optimizer seed before method comparison and bootstrap the ten paired seed effects 10,000 times with seed 20260811. A strictly positive 95% interval is described as supported, a strictly negative interval as unsupported, and an interval containing zero as mixed. These are descriptive labels, not a publication gate. Every one of the 12 cells remains visible.

For the representative $n = 19, s = 0.50$ cell, no-condition minus conditioned training profile TVD is 0.2668 [0.2587, 0.2737]; held-out target-aware adjacent-reference minus conditioned profile TVD is 0.0016 [-0.0012, 0.0043]; and held-out coordinate-weighted mixture minus conditioned profile TVD is -0.0029 [-0.0053, -0.0006]. The conditioned-minus-per-condition-reference training fidelity gap is 0.0878 [0.0860, 0.0897] (positive values mean a cost of sharing). The corresponding raw-bitstring adjacent-reference effect is -0.0606 [-0.0749, -0.0461]. Across the full surface, conditioning improves the training-condition Hamming-weight profiles over the no-condition model in all 12 size–width settings. At unseen conditions, separately trained neighboring circuits are usually more accurate: their target-aware oracle is better in ten settings and statistically indistinguishable in two. The coordinate-weighted mixture is better in nine settings, indistinguishable in one, and worse in two. Thus the experiment shows robust conditional control and interpolation relative to no conditioning, together with a measurable fidelity cost for sharing one circuit across conditions.

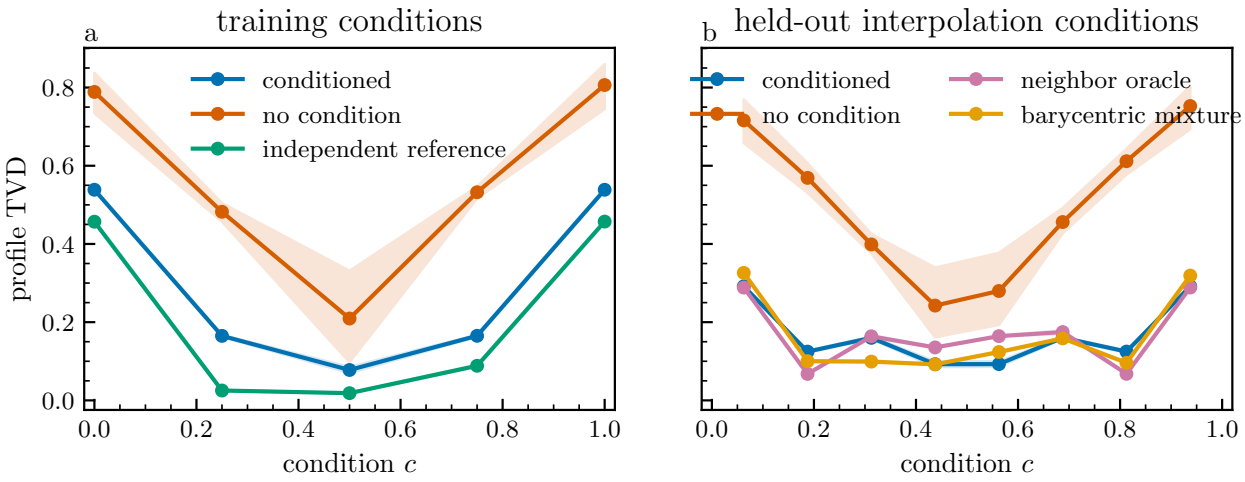


Figure 2: Training-condition control and held-out interpolation at $n = 19, s = 0.50$. Lines are ten-seed means and bands are deterministic 95% seed-bootstrap intervals. The held-out panel compares the conditioned circuit with no conditioning, a target-aware choice between the two adjacent per-condition circuits, and their coordinate-weighted mixture.

8 Performance across size, width, and peak location

Figure 3 resolves the aggregate statements into the complete 4×3 performance surface. Positive and negative cells use the same color scale, and the numerical paired effect is printed in each cell. The per-condition curves and exhaustive intervals in the supplement reveal whether an aggregate seed mean masks a difficult peak location. In particular, the endpoints probe truncation asymmetry, while the common odd-sixteenth grid makes held-out comparisons identical across all training grids.

The width sweep changes two distinct difficulties. Narrow profiles demand accurate concentration in weight, while wider profiles allocate mass to more shells. Increasing n also increases both the output space and the complete two-local parameter count. We therefore avoid pooling cells across n or s into one headline effect. The heatmaps summarize direction and magnitude; the table in the supplement retains the paired interval and cell-level wins.

9 Condition-count and data-allocation scaling

The K study asks whether adding training conditions helps because the model sees the conditional curve more densely or merely because it receives more data. The fixed-per-condition regime increases total samples with K; the fixed-total regime holds total samples constant. Because all subsets are prefixes of the same target-specific master pools, differences cannot be attributed to unrelated draws. We report the per-condition and worst- condition profile TVD rather than only the overall mean.

The shared circuit always has 190 phases and 2,000 optimizer updates. An per-condition reference has $190K$ stored phases and $2,000K$ optimizer updates. Its larger optimization work is part of the baseline definition, not hidden from the comparison. The nine-circuit per-condition reference is particularly generous: its nested outputs are reused for all K values, whereas shared circuits are retrained for their declared grid and data regime.

10 Temperature control in the Curie–Weiss model

The Gaussian task isolates conditional translation; Curie–Weiss tests whether the same fixed-circuit mechanism can follow a family that changes qualitatively with its condition. For spins $s_i \in \{-1, +1\}$ and $M = \sum_i s_i$, we use

$$H(s) = -\frac{M^2}{2n}, \qquad p_\beta(s) = \frac{e^{-\beta H(s)}}{Z_\beta}. \tag{9}$$

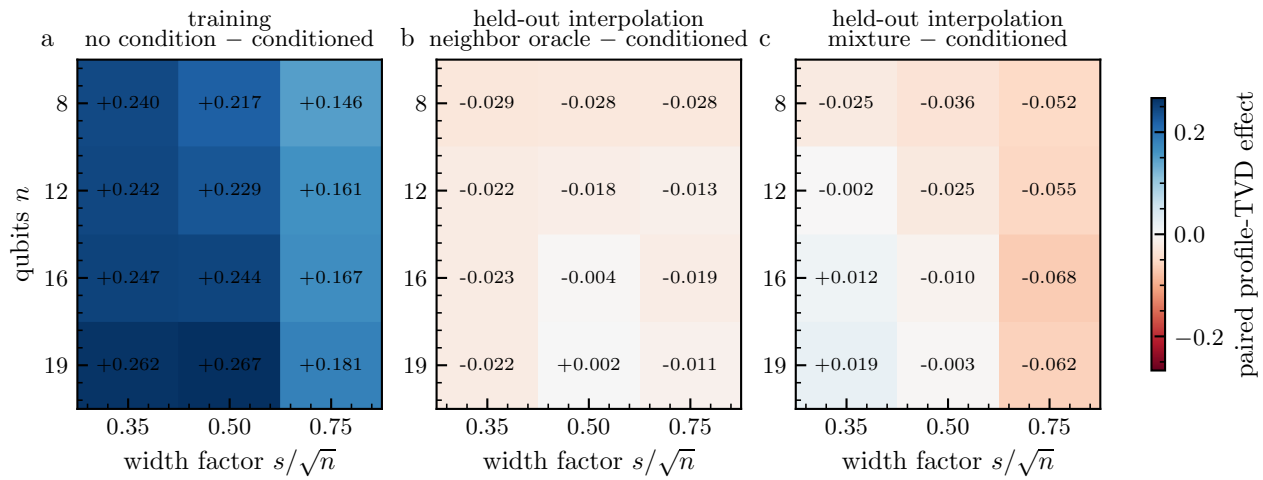


Figure 3: Complete Gaussian performance surface. Each entry is a ten-seed paired profile-TVD effect: no-condition minus conditioned at training conditions, and adjacent-reference oracle or coordinate-weighted mixture minus conditioned at held-out conditions. Positive values favor the conditioned circuit.

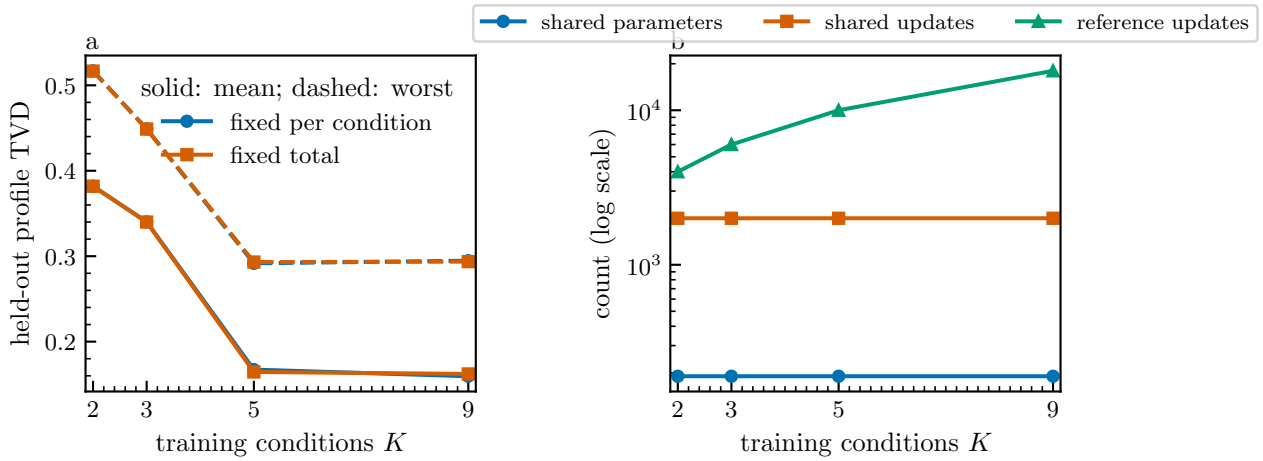


Figure 4: Condition-count characterization at $n = 19, s = 0.50$. (a) Mean and worst-condition held-out profile TVD for the fixed-per-condition and fixed-total regimes. (b) Shared-model parameter count and optimization work compared with independently trained references.

At zero field the law is invariant under spin flip and permutation. Raising β moves its finite-size magnetization profile from a disordered central distribution, through enhanced fluctuations near the thermodynamic critical value $\beta_c = 1$, toward an ordered bimodal distribution. Magnetization-profile TVD is consequently the primary metric; raw bitstring TVD and the thermodynamic response functions test structure hidden by that quotient.

We retain the Gaussian encoder without adapting it to Curie–Weiss outcomes: $c = \beta - 0.5$ and $\alpha(\beta) = \pi[0.35 - 0.70c]$. Six training temperatures $\{0.5, 0.7, 0.9, 1.1, 1.3, 1.5\}$ alternate with five unseen midpoints $\{0.6, 0.8, 1.0, 1.2, 1.4\}$ at $n \in \{8, 12, 16\}$. In particular, $\beta = 1$ is never available during optimization or checkpoint selection. The symmetry-aligned model learns one coefficient shared by all one-body phases and one shared by every two-body phase. These coefficients expand over the complete $n(n+1)/2$ generator sequence, so parameter sharing is an inductive bias rather than a reduced topology. The unrestricted conditioned circuit supplies an ablation, while a matched no-condition model and six independently trained temperature references supply the principal baselines.

All four model families use the same 2,000-update finite-sample Conditional MMD protocol as the Gaussian study. Ten matched seeds pro-

duce 270 atomic fits. Every frozen model is enumerated at all eleven temperatures; held-out references are compared both through a target-aware choice of the better adjacent model and through their target-blind equal mixture. Conditions are averaged within seed before 10,000 deterministic paired bootstrap resamples, and sizes are never pooled. We impose no empirical pass threshold.

Figure 5 shows how the learned profiles evolve from the hot endpoint through the unseen critical point to the cold endpoint. Conditioning has a clear but size-dependent effect. Relative to the matched no-condition control, mean profile TVD improves at both training and held-out temperatures for $n = 12$ and 16: the held-out improvements are $0.142\,[0.133, 0.152]$ and $0.143\,[0.141, 0.144]$, respectively. At $n = 8$, the effect reverses, $-0.027\,[-0.045, -0.005]$. At the unseen critical point, the comparison is favorable for $n = 12$, inconclusive for $n = 16$, and adverse for $n = 8$.

The stronger interpolation references expose the remaining fidelity gap. The conditioned model loses to both the adjacent-model oracle and the target-blind equal mixture at every size; against the mixture, the held-out effects are -0.158, -0.176, and -0.224 for $n = 8, 12, 16$. Nor does the two-coefficient physical bias improve fidelity: unrestricted phases reduce held-out profile TVD by 0.137–0.162 across the three sizes. The tied model therefore demonstrates temperature response, not a superior interpolator or thermal-state generator. Absolute errors remain substantial near $\beta = 1$ and at $\beta = 1.5$, as Figs. 5 and 6 show.

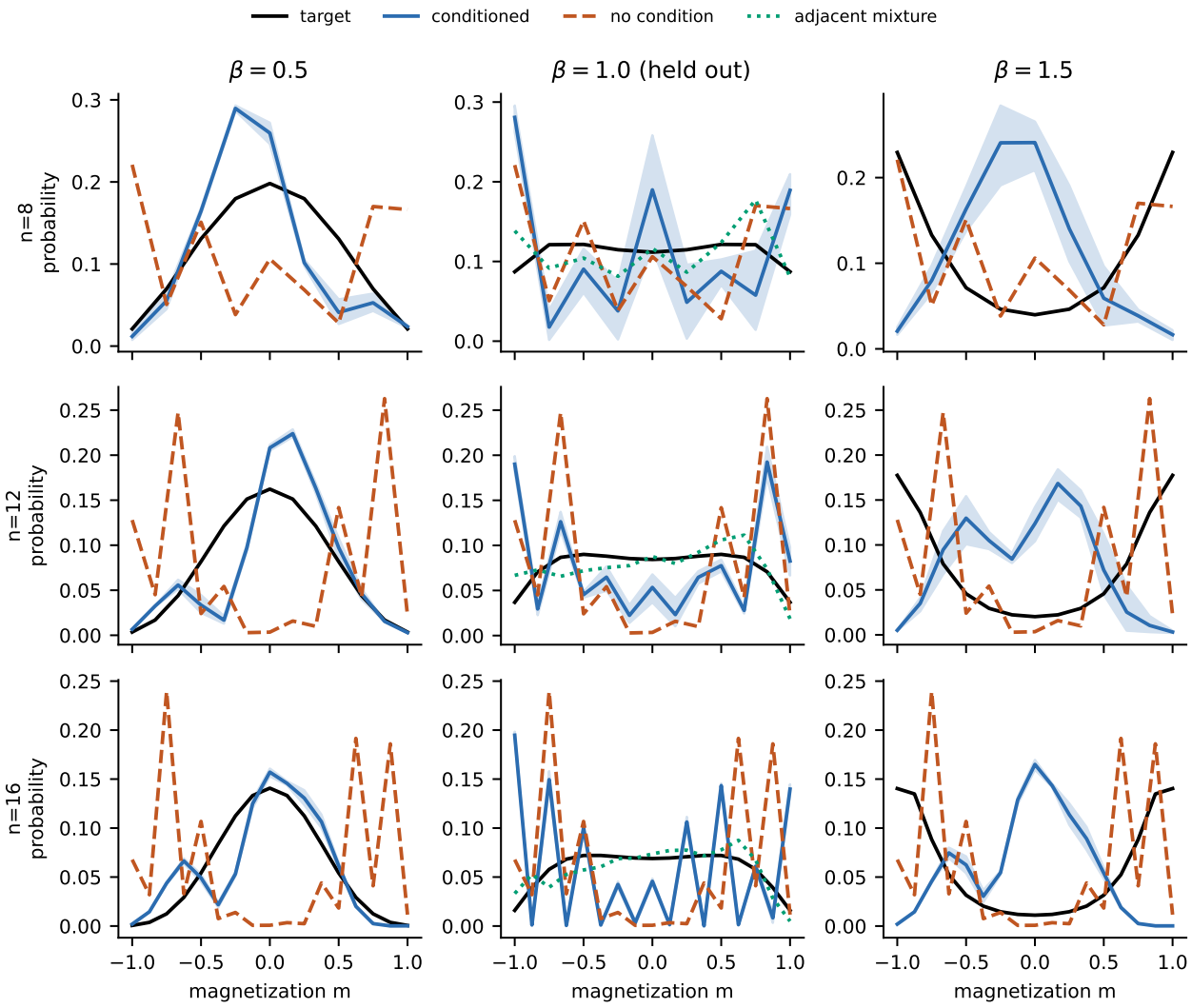


Figure 5: Temperature-controlled Curie–Weiss magnetization profiles at $n = 8, 12, 16$. Columns show the hot endpoint, the unseen critical interpolation point, and the cold endpoint. Target profiles are black; conditioned curves and 95% bootstrap bands summarize ten matched seeds. The no-condition and target-blind adjacent-mixture references expose whether the response comes from the supplied temperature.

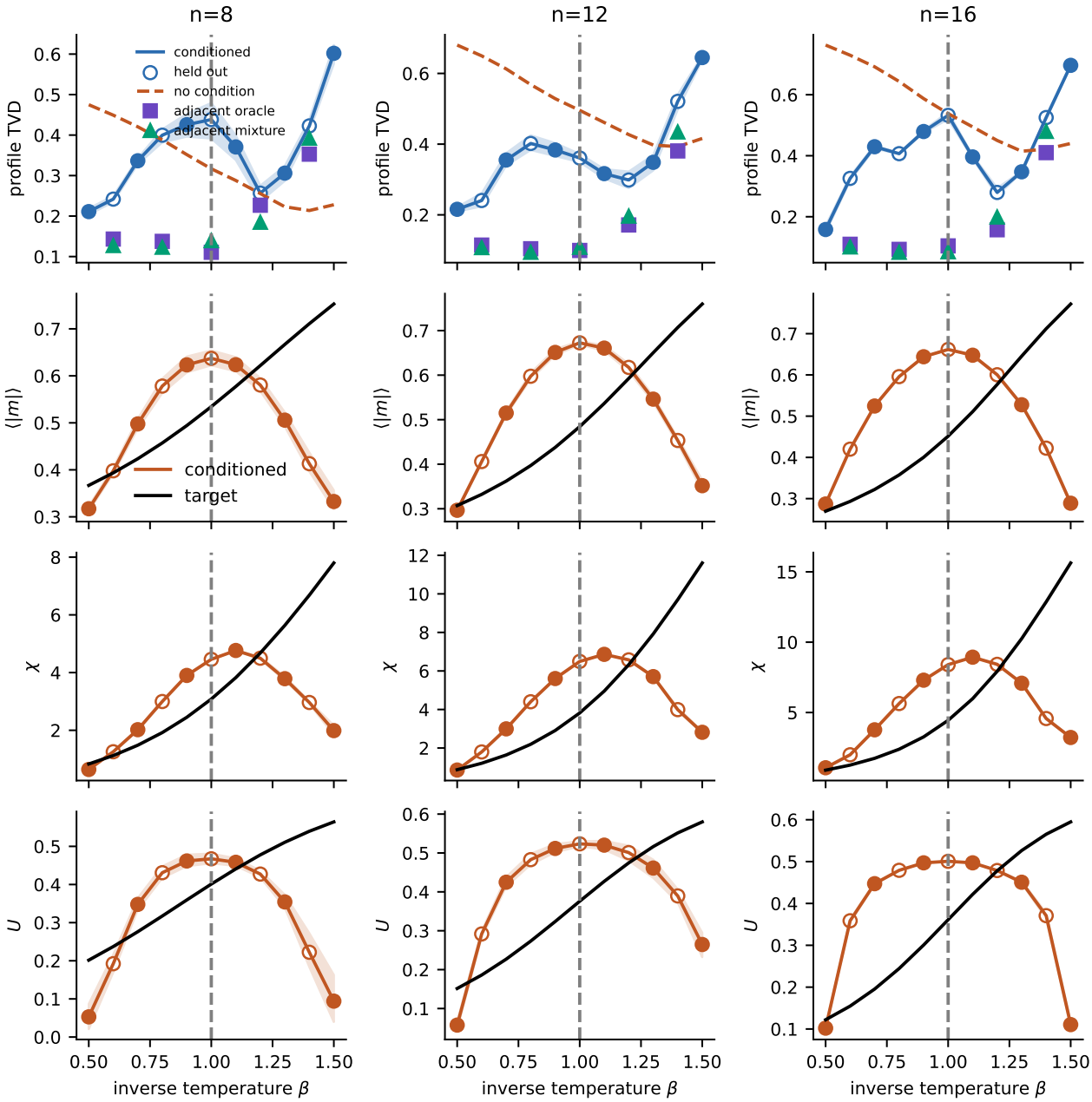


Figure 6: Interpolation fidelity and physical response. Magnetization-profile TVD distinguishes training and held-out temperatures and compares the conditioned circuit with no condition, the adjacent-reference oracle, and the equal adjacent mixture. Absolute magnetization, susceptibility, and Binder cumulant are shown against the exact target; filled and hollow markers denote training and held-out temperatures, and the dashed line marks the unseen $\beta = 1$ point.

11 Compact Gaussian ablations

Three archived studies test modeling choices outside the frozen factorial. First, complete two-local connectivity is compared with lattice and heavy-hex generator banks on a two-peak 19-qubit task. Second, the broadcast-angle map is compared with a symmetric-reupload encoding at matched parameter count. Third, ancilla-assisted models are compared with visible-only circuits whose additional higher-order generators match the parameter counts. Figure 7 reports these studies descriptively and separately from the 1,020-model campaign because their target grids and training protocols are not identical.

The ancilla comparison is especially easy to overinterpret. Adding ancillas changes the effective visible marginal family, whereas adding visible-only generators changes capacity without changing the measured subsystem. A difference between them is evidence about this benchmark and construction, not a general ancilla advantage. Target-symmetry-aware post-processing is analyzed separately in the supplement and is excluded from every raw-circuit claim.

12 Limitations, cost, and reproducibility

Both target families are one-dimensional and exchangeable. Low profile TVD means the model places the right total mass in each Hamming shell; it does not mean that probability is uniform among strings inside a shell. Raw bitstring TVD is therefore reported in every cell and is the main representational limitation. The benchmark demonstrates interpolation inside a fixed interval, not extrapolation, robustness to shifted target families, or universal conditional generation.

For the symmetry-tied Curie–Weiss circuit, the output is itself permutation symmetric. Its shell probabilities therefore determine the full bitstring law, and its profile and raw bitstring TVDs agree up to numerical precision. This equality does not hold for the unrestricted ablation or for the Gaussian circuits and is not evidence for general bitstring-level fidelity.

Our operational classical-trainability statement is likewise bounded. The training objective and gradients use finite classical estimators, but exact post-selection evaluation enumerates 2^n probabilities. Runtime grows with system size, connectivity, estimator budget, and optimizer work. Nothing here establishes efficient classical sampling of the trained output, a quantum advantage, or a favorable asymptotic optimization landscape.

Curie–Weiss is a controlled finite-size probability benchmark, not a claim of state-of-the-art thermal-state preparation or exact critical scaling. Parameter sharing deliberately builds known exchange symmetry into the model, and the isotropic Hamming kernel is aligned with the magnetization profile but does not resolve spatial correlations. General many-body Hamiltonians, extrapolation in temperature, and nonexchangeable physical observables remain outside the evidence.

The campaign is manifest-driven and immutable. It declares exactly 1,020 run identities, rejects undeclared or duplicate completion, binds datasets and estimator streams by role, and refuses partial reference-collection reuse. Execution is resumable and shardable. Each result binds the manifest, source tree, environment, revision, dataset, checkpoint, probability archive, invocation, runtime, and peak memory. Figures, tables, and claims are generated only from the complete report. The supplement describes regeneration, archive checks, and clean-build verification.

13 Conclusion

Product-state conditioning gives a precise fixed-deployment route to conditional IQP generation: one compiled commuting circuit and one phase vector respond to a classical condition through local input rotations. The selected Hamming-kernel training objective and gradients remain accessible to classical probabilistic estimation, while full-output access is not claimed. Across the Gaussian campaign, conditioning improves training-condition profile TVD in all 12 size–width cells. Held-out outputs, however, usually do not outperform the generous adjacent-reference oracle or its target-blind barycentric mixture: the adjacent-reference comparison is mixed in two cells and unsupported in ten, while the mixture comparison is supported in two, mixed in one, and unsupported in nine. The result is therefore evidence for robust conditional

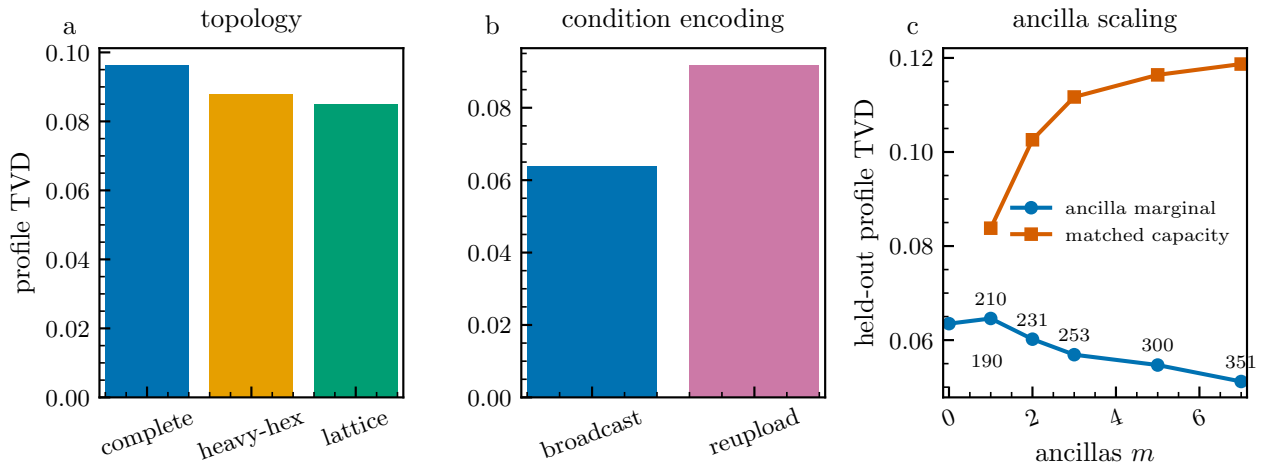


Figure 7: Compact archived Gaussian ablations. (a) Complete, lattice, and heavy-hex topologies. (b) Broadcast-angle and symmetric- reupload encodings. (c) Ancilla scaling; annotations give parameter counts. Protocols differ from the frozen factorial, so panels are descriptive rather than pooled campaign evidence.

control, not a broad claim of superior interpolation. This positive-and-negative performance map provides a transparent foundation for improving fixed-circuit conditional models while keeping the present claim restricted to the tested Gaussian Hamming-profile family. The Curie–Weiss study shows that the same product-state mechanism can produce a physically interpretable temperature response: it beats no conditioning at 12 and 16 spins, but not at 8. It does not beat adjacent-temperature references, and the unrestricted ablation shows that the proposed two-coefficient symmetry sharing is too restrictive here. These negative comparisons and the critical/cold-regime errors delimit the result: this is a conditional-generation showcase, not evidence for general thermal-state generation.

Data and code availability

The source repository contains the frozen manifest, role-tagged datasets, immutable ledgers, checkpoints, exact probability archives, report, claim registry, figure/table generators, and clean-build scripts. The canonical PDF is promoted only after all declared identities and evaluations complete.

Release metadata

Author names, affiliations, individual contributions, competing interests, arXiv category and license, and final human proof approval remain to be supplied and verified by the research team. This document is not upload-ready. No upload or submission is automated.

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A Analytical details

This appendix derives the response and estimator statements used in the main text. They specialize established circuit-Fourier and probabilistic-IQP results to product-state-conditioned notation and are not presented as novel central theorems.

A.1 Full-probability response support

One encoded qubit has density operator

$$\begin{aligned}\rho_i(\alpha_i) &= R_Y(\alpha_i)|0\rangle\langle 0|R_Y(\alpha_i)^\dagger \\ &= \tfrac{1}{2}\left(I + \sin\alpha_i X + \cos\alpha_i Z\right).\end{aligned} \tag{10}$$

Each entry is a linear combination of angle frequencies $-1, 0, +1$. Tensoring and substituting $\alpha_i(c) = b_i + w_i^\mathsf{T} c$ gives fixed matrices A_s such that

$$\rho(c) = \sum_{s\in\{-1,0,1\}^n} A_s e^{i\sum_i s_i b_i} e^{i(\sum_i s_i w_i)^\mathsf{T} c}. \tag{11}$$

For fixed unitary U and outcome x, $q(x \mid c) = \mathrm{Tr}[U^\dagger|x\rangle\langle x|U\rho(c)]$ is a fixed linear functional of $\rho(c)$ and introduces no new condition frequency. This proves Eq. (7). Zero rows, coincident sums, and coefficient cancellations can only shrink the support. The trace coefficient can still require exponential work to evaluate.

For the broadcast scalar encoding in Eq. (3), every row of W is identical. Frequencies therefore collapse to integer multiples of the single slope, with multiplicities generated by different sign patterns. This degeneracy is useful for a one-coordinate location target but also limits the independent conditional directions available at fixed width.

A.2 Pauli observable filter

The final Hadamards map measured Z_a to X_a. A diagonal generator Z_{g_j} satisfying $a\cdot g_j = 0 \pmod 2$ commutes with X_a and cancels. For an active generator,

$$e^{-i\theta_j Z_{g_j}} X_a e^{i\theta_j Z_{g_j}} = X_a e^{2i\theta_j Z_{g_j}}. \tag{12}$$

Since the diagonal generators commute,

$$\begin{aligned}D_Z(\theta)^\dagger X_a D_Z(\theta) = X_a \prod_{j:a\cdot g_j=1} &[\cos(2\theta_j) I \\ &+ i\sin(2\theta_j) Z_{g_j}].\end{aligned} \tag{13}$$

Expanding groups active subsets by $h = \bigoplus_{j\in S} g_j$. Conjugating $X_a Z_h$ through the first Hadamards produces local $Z_a X_h$ factors. On the real product input,

$$\langle X_i\rangle = \sin\alpha_i, \qquad \langle Z_i\rangle = \cos\alpha_i, \qquad \langle Y_i\rangle = 0. \tag{14}$$

Every term with $h\wedge a \neq 0$ therefore vanishes. Surviving terms factor into sines and cosines on $\mathrm{supp}(a)\cup\mathrm{supp}(h)$; expanding into exponentials yields the signed sums stated in the proposition.

A.3 Hamming-kernel mask identity and estimator

For one bit, let $a \sim \mathrm{Bernoulli}(q)$. Then

$$\mathbb{E}_a[(-1)^{a(x-y)}] = \begin{cases} 1, & x = y, \\ 1-2q, & x \neq y. \end{cases} \tag{15}$$

Choosing $q = q_\sigma$ from Eq. (5) gives $1-2q = e^{-1/(2\sigma^2)}$. Independence across bits yields Eq. (6). Let $\phi_a(x) = (-1)^{a\cdot x}$. For target samples $y_1,\ldots,y_T$ and model samples $x_1,\ldots,x_R$, the fixed-mask unbiased squared

discrepancy is

$$\widehat{\Delta}_a^2 = \frac{1}{R(R-1)} \sum_{r \neq r'} \phi_a(x_r)\phi_a(x_{r'}) + \frac{1}{T(T-1)} \sum_{t \neq t'} \phi_a(y_t)\phi_a(y_{t'}) - \frac{2}{RT} \sum_{r,t} \phi_a(x_r)\phi_a(y_t). \tag{16}$$

Averaging over masks, bandwidths, and conditions gives sampled Conditional MMD. Differentiation acts only on the model expectation because target and mask laws do not depend on phase parameters. This gives estimator access, not a lower bound on gradient magnitude or a convergence guarantee.

After checkpoint selection, the exact population value is computed from Walsh–Hadamard coefficients. With $\widehat{p}(a) = \sum_x p(x)(-1)^{a \cdot x}$,

$$\mathrm{MMD}_\sigma^2(p,q) = \sum_{a \in \{0,1\}^n} q_\sigma^{|a|}(1-q_\sigma)^{n-|a|} \left[\widehat{p}(a) - \widehat{q}(a)\right]^2. \tag{17}$$

B Target construction and data isolation

For a nominal location ℓ and width $s\sqrt{n}$, define unnormalized shell mass

$$r_k(\ell, s) = \exp\left[-\frac{(k-\ell)^2}{2s^2 n}\right], \qquad k = 0, \ldots, n. \tag{18}$$

Truncation changes the realized mean, so a monotone scalar root solve chooses ℓ such that $\sum_k k r_k / \sum_k r_k = \mu_n(c)$. The resulting normalized shell mass $P(k \mid c)$ induces

$$p(x \mid c) = \frac{P(|x| \mid c)}{\binom{n}{|x|}}. \tag{19}$$

This is an exact analytic law. Training nonetheless sees only finite sampled strings.

Each combination of n, width factor, and master seed has one immutable data archive. It stores the nine-condition master grid, role-tagged train and validation strings, derived random seeds, and hashes. The main five-condition grid selects even master indices. The $K = 2, 3, 5$ grids are nested subsets of the nine-grid conditions. Fixed-total arrays are exact prefixes of the same 10,000/2,000 per-target masters. The archive schema has no held-out role, and the loader rejects any such role before training.

C Curie–Weiss target, observables, and protocol

For a bitstring x, write $s_i = 2x_i - 1$, $M = \sum_i s_i$, and $m = M/n$. The exact target law is

$$p_\beta(x) = Z_\beta^{-1} \exp\left(\frac{\beta M^2}{2n}\right). \tag{20}$$

It depends only on Hamming weight, so all strings in one magnetization shell have equal probability. Spin flip maps M to $-M$ and leaves the law unchanged. These identities are checked directly before any sample archive is written.

Exact post-selection evaluation records profile and raw bitstring TVD and

$$\chi = \beta n\big(\langle m^2 \rangle - \langle m \rangle^2\big), \tag{21}$$

$$U = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}, \tag{22}$$

$$C/n = \frac{\beta^2}{n} \mathrm{Var}(H), \tag{23}$$

together with $\langle |m| \rangle$, energy per spin, and exact multibandwidth MMD. At finite n, $\beta = 1$ is a crossover probe, not an estimate of an asymptotic critical exponent.

Table 1: Curie–Weiss paired profile-TVD effects. Positive values favor the symmetry-aligned conditioned circuit. The final column is unrestricted minus aligned, so a negative value favors unrestricted phases. Intervals are paired 95% bootstrap intervals over the ten matched seeds.

n	train: no cond.	held: no cond.	held: oracle	held: mixture	unrestricted − aligned
8	-0.044 [-0.063, -0.022]	-0.027 [-0.045, -0.005]	-0.158 [-0.173, -0.142]	-0.158 [-0.174, -0.141]	-0.137 [-0.171, -0.102]
12	+0.139 [+0.133, +0.144]	+0.142 [+0.133, +0.152]	-0.191 [-0.200, -0.181]	-0.176 [-0.186, -0.166]	-0.145 [-0.176, -0.116]
16	+0.147 [+0.145, +0.150]	+0.143 [+0.141, +0.144]	-0.239 [-0.243, -0.235]	-0.224 [-0.228, -0.221]	-0.162 [-0.185, -0.134]

One role-separated archive is generated for each size and matched seed. It contains 10,000 training and 2,000 validation strings at each of six training temperatures. Training and validation use independent hash-derived streams. The schema has neither held-out samples nor held-out seed fields. Exact laws at the five midpoints become accessible only after checkpoint selection.

The exchange-symmetric expansion has one column for all one-body generators and one for all two-body generators. Multiplying it by two learned coefficients produces the phase vector consumed by the unchanged complete generator sequence. The unrestricted ablation replaces this matrix by the identity. It changes parameter sharing but not generator order, connectivity, or condition encoder.

The archived direct-TVD global search is a capacity ceiling only. It uses exact probabilities and a non-scalable target objective, so it is excluded from headline comparisons and from the finite-sample MMD training claim.

Table 1 gives every aggregate paired effect, while Table 2 gives the complete temperature grid. The latter reports model/target pairs for physical observables. Because the tied model is permutation symmetric, profile and raw bitstring TVD coincide to displayed precision. The underlying report retains every seed row and every exact profile; the 270 probability archives retain the full 2^n laws.

Table 2: Complete Curie–Weiss response surface for the symmetry-aligned conditioned circuit. Role T is training and H is held out. Physical-response entries are model/target means over ten seeds.

n	β	role	profile	raw	$\langle\lvert m\rvert\rangle$	χ	U	E/n	C/n
8	0.5	T	0.211	0.211	0.317/0.367	0.64/0.83	0.053/0.202	-0.086/-0.104	0.027/0.030
8	0.6	H	0.242	0.242	0.398/0.394	1.26/1.12	0.192/0.237	-0.132/-0.117	0.072/0.051
8	0.7	T	0.337	0.337	0.498/0.424	2.02/1.48	0.348/0.276	-0.191/-0.132	0.137/0.080
8	0.8	H	0.400	0.400	0.578/0.457	3.00/1.92	0.431/0.317	-0.241/-0.150	0.210/0.121
8	0.9	T	0.426	0.426	0.623/0.494	3.90/2.45	0.462/0.359	-0.271/-0.170	0.293/0.173
8	1.0	H	0.438	0.438	0.637/0.534	4.46/3.08	0.467/0.401	-0.281/-0.193	0.378/0.237
8	1.1	T	0.371	0.371	0.624/0.577	4.77/3.83	0.458/0.441	-0.271/-0.217	0.444/0.310
8	1.2	H	0.257	0.257	0.580/0.622	4.50/4.68	0.427/0.478	-0.240/-0.244	0.478/0.388
8	1.3	T	0.306	0.306	0.506/0.667	3.79/5.64	0.354/0.511	-0.192/-0.271	0.467/0.464
8	1.4	H	0.423	0.423	0.413/0.711	2.96/6.68	0.222/0.540	-0.137/-0.298	0.385/0.532
8	1.5	T	0.602	0.602	0.333/0.753	1.99/7.80	0.094/0.564	-0.092/-0.325	0.252/0.585
12	0.5	T	0.216	0.216	0.297/0.307	0.86/0.88	0.057/0.152	-0.075/-0.073	0.031/0.025
12	0.6	H	0.241	0.241	0.406/0.332	1.80/1.21	0.292/0.186	-0.128/-0.084	0.080/0.044
12	0.7	T	0.355	0.355	0.515/0.362	2.99/1.64	0.425/0.227	-0.184/-0.098	0.144/0.074
12	0.8	H	0.402	0.402	0.597/0.397	4.40/2.19	0.483/0.273	-0.229/-0.114	0.223/0.118
12	0.9	T	0.383	0.383	0.651/0.438	5.61/2.91	0.512/0.324	-0.260/-0.135	0.305/0.181
12	1.0	H	0.361	0.361	0.673/0.484	6.49/3.82	0.523/0.376	-0.271/-0.159	0.378/0.265
12	1.1	T	0.317	0.317	0.661/0.536	6.86/4.95	0.520/0.427	-0.260/-0.188	0.431/0.368
12	1.2	H	0.299	0.299	0.617/0.593	6.58/6.32	0.501/0.475	-0.229/-0.220	0.452/0.480
12	1.3	T	0.349	0.349	0.546/0.650	5.71/7.92	0.461/0.516	-0.185/-0.254	0.426/0.589
12	1.4	H	0.521	0.521	0.453/0.707	3.99/9.70	0.390/0.552	-0.135/-0.289	0.356/0.677
12	1.5	T	0.645	0.645	0.352/0.760	2.82/11.60	0.265/0.580	-0.089/-0.322	0.255/0.730
16	0.5	T	0.158	0.158	0.287/0.269	1.07/0.90	0.102/0.122	-0.067/-0.056	0.031/0.021
16	0.6	H	0.326	0.326	0.420/0.293	2.00/1.26	0.359/0.155	-0.125/-0.066	0.083/0.038
16	0.7	T	0.429	0.429	0.525/0.322	3.77/1.74	0.447/0.195	-0.178/-0.078	0.164/0.067
16	0.8	H	0.407	0.407	0.596/0.357	5.64/2.39	0.479/0.244	-0.223/-0.093	0.287/0.113
16	0.9	T	0.479	0.479	0.644/0.400	7.30/3.27	0.497/0.301	-0.254/-0.113	0.427/0.183
16	1.0	H	0.532	0.532	0.662/0.451	8.40/4.44	0.501/0.361	-0.266/-0.139	0.562/0.283
16	1.1	T	0.396	0.396	0.648/0.511	8.93/5.98	0.497/0.422	-0.254/-0.170	0.638/0.411
16	1.2	H	0.280	0.280	0.600/0.577	8.42/7.93	0.479/0.478	-0.223/-0.206	0.647/0.556
16	1.3	T	0.347	0.347	0.528/0.645	7.08/10.25	0.451/0.526	-0.178/-0.246	0.558/0.691

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Table 3: Exact population MMD^2 by individual Hamming-kernel bandwidth, averaged over temperatures and matched seeds. The training objective averages these three bandwidths.

n	role	$\sigma = 1$	$\sigma = 2$	$\sigma = 4$
8	training	0.02705	0.00824	0.00171
8	held out	0.01882	0.00560	0.00118
12	training	0.02886	0.01153	0.00208
12	held out	0.02171	0.00928	0.00194
16	training	0.03212	0.01796	0.00332
16	held out	0.02774	0.01917	0.00554

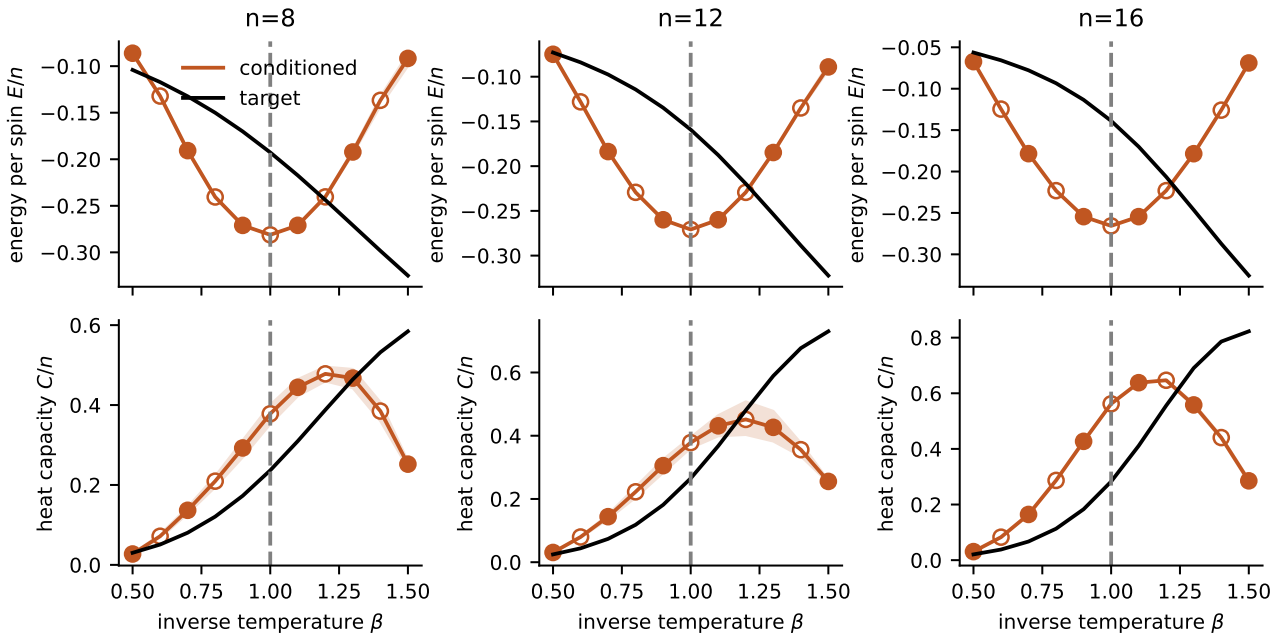


Figure 8: Energy and heat-capacity response of the symmetry-aligned conditioned circuit. Filled and hollow markers denote training and held-out temperatures; bands are 95% paired-bootstrap intervals over ten seeds. The vertical line marks the unseen $\beta = 1$ point.

n	β	role	profile	raw	$\langle\lvert m\rvert\rangle$	χ	U	E/n	C/n
16	1.4	H	0.525	0.525	0.422/0.712	4.57/12.86	0.371/0.565	-0.126/-0.287	0.441/0.786
16	1.5	T	0.696	0.696	0.289/0.772	3.22/15.63	0.111/0.595	-0.069/-0.326	0.285/0.823

D Complete factorial results

Table 5 reports every size–width cell. Effects average conditions within seed and then pair by seed. Positive values favor the conditioned model. Status letters are S (95% interval strictly positive), M (contains zero), and U (strictly negative), ordered as training control, held-out adjacent-reference oracle, and held-out coordinate-weighted mixture. They describe the observed uncertainty and do not implement a success rule.

For auditability, the machine-readable report also records cell-level wins at each seed–condition pair, mean and width error, exact Conditional MMD, checkpoint selection, and model and target shell probabilities. The complete probability artifact retains all 2^n model probabilities at every declared evaluation coordinate.

Table 4: Frozen data allocation in the condition-count study.

K	fixed per condition		fixed total	
	train	validation	train	validation
2	10,000	2,000	9,000	1,800
3	10,000	2,000	6,000	1,200
5	10,000	2,000	3,600	720
9	10,000	2,000	2,000	400

Table 5: All factorial profile-TVD effects with deterministic paired 95% bootstrap intervals.

n	$s/\sqrt{n}$	no condition − conditioned	adjacent-reference oracle − conditioned	mixture − conditioned	S/M/U
8	0.35	+0.2402 [+0.2278, +0.2536]	-0.0292 [-0.0468, -0.0108]	-0.0248 [-0.0418, -0.0064]	S/U/U
8	0.50	+0.2170 [+0.2015, +0.2418]	-0.0280 [-0.0336, -0.0225]	-0.0363 [-0.0432, -0.0292]	S/U/U
8	0.75	+0.1460 [+0.1293, +0.1714]	-0.0276 [-0.0339, -0.0214]	-0.0525 [-0.0584, -0.0461]	S/U/U
12	0.35	+0.2419 [+0.2352, +0.2505]	-0.0221 [-0.0255, -0.0190]	-0.0016 [-0.0052, +0.0019]	S/U/M
12	0.50	+0.2291 [+0.2213, +0.2391]	-0.0180 [-0.0213, -0.0146]	-0.0251 [-0.0283, -0.0216]	S/U/U
12	0.75	+0.1608 [+0.1515, +0.1722]	-0.0129 [-0.0151, -0.0107]	-0.0548 [-0.0567, -0.0526]	S/U/U
16	0.35	+0.2472 [+0.2421, +0.2519]	-0.0229 [-0.0279, -0.0174]	+0.0117 [+0.0067, +0.0172]	S/U/S
16	0.50	+0.2440 [+0.2354, +0.2537]	-0.0040 [-0.0071, +0.0001]	-0.0099 [-0.0133, -0.0057]	S/M/U
16	0.75	+0.1667 [+0.1570, +0.1770]	-0.0192 [-0.0205, -0.0178]	-0.0681 [-0.0696, -0.0666]	S/U/U
19	0.35	+0.2620 [+0.2594, +0.2645]	-0.0217 [-0.0240, -0.0192]	+0.0188 [+0.0164, +0.0213]	S/U/S
19	0.50	+0.2668 [+0.2587, +0.2737]	+0.0016 [-0.0012, +0.0043]	-0.0029 [-0.0053, -0.0006]	S/M/U
19	0.75	+0.1807 [+0.1648, +0.1999]	-0.0111 [-0.0165, -0.0049]	-0.0619 [-0.0670, -0.0560]	S/U/U

E Raw-bitstring diagnostics

Table 7 keeps the raw-law limitation adjacent to the primary profile result. Because targets are uniform within shells, profile TVD is a quotient metric: it ignores redistribution within a shell. The difference between profile and bitstring TVD quantifies that unmodeled structure. A positive paired effect still means the conditioned model is closer than the named baseline, but a large absolute TVD remains a limitation even when the relative comparison is favorable.

F Condition-count results

Table 9 reports both data regimes, the average and worst held-out coordinate, parameter count, and optimization work. The shared-model parameter and update counts do not grow with K; the independent reference collection's stored parameters and optimizer work grow linearly. The per-condition JSON rows behind this table are retained so that a mean cannot hide a single difficult coordinate.

Table 11 exposes every held-out coordinate rather than leaving those values only in the machine-readable report. Entries are ten-seed means of profile TVD; lower is better.

Table 11: Per-condition held-out profile TVD in the condition-count study.

Table 6: Conditioned-model profile-TVD wins at the seed–Condition level in every factorial cell. Denominators are 50 Training comparisons and 80 Held-out comparisons per cell.

n	$s/\sqrt{n}$	vs. no condition	vs. adjacent-reference oracle	vs. mixture
8	0.35	50/50	23/80	28/80
8	0.50	50/50	8/80	10/80
8	0.75	50/50	19/80	6/80
12	0.35	48/50	30/80	39/80
12	0.50	48/50	18/80	21/80
12	0.75	50/50	20/80	20/80
16	0.35	46/50	11/80	50/80
16	0.50	48/50	38/80	30/80
16	0.75	50/50	20/80	20/80
19	0.35	46/50	27/80	58/80
19	0.50	47/50	40/80	38/80
19	0.75	49/50	28/80	20/80

Table 7: Raw bitstring TVD across all factorial cells. The middle columns are the conditioned model's absolute mean TVD; effects are paired 95% intervals.

n	$s/\sqrt{n}$	training	held out	no condition − conditioned	adjacent-reference oracle − conditioned
8	0.35	0.3394	0.2660	+0.2388 [+0.2232, +0.2554]	-0.0243 [-0.0445, -0.0054]
8	0.50	0.3086	0.2371	+0.2055 [+0.1855, +0.2306]	-0.0380 [-0.0491, -0.0277]
8	0.75	0.3229	0.2586	+0.1288 [+0.1018, +0.1699]	-0.0099 [-0.0165, -0.0032]
12	0.35	0.4179	0.3274	+0.2578 [+0.2494, +0.2684]	-0.0371 [-0.0466, -0.0271]
12	0.50	0.3661	0.2825	+0.2594 [+0.2388, +0.2782]	-0.0356 [-0.0470, -0.0249]
12	0.75	0.3596	0.2954	+0.1687 [+0.1409, +0.1999]	+0.0156 [+0.0083, +0.0233]
16	0.35	0.4618	0.3587	+0.2515 [+0.2406, +0.2632]	-0.0429 [-0.0538, -0.0319]
16	0.50	0.4061	0.3135	+0.2575 [+0.2402, +0.2765]	-0.0418 [-0.0510, -0.0318]
16	0.75	0.3929	0.3279	+0.2100 [+0.1710, +0.2486]	+0.0250 [+0.0190, +0.0313]
19	0.35	0.4964	0.3905	+0.2394 [+0.2256, +0.2523]	-0.0610 [-0.0776, -0.0458]
19	0.50	0.4380	0.3444	+0.2559 [+0.2370, +0.2723]	-0.0606 [-0.0749, -0.0461]
19	0.75	0.4292	0.3573	+0.1940 [+0.1476, +0.2405]	+0.0074 [-0.0032, +0.0165]

K	Regime	c	Conditioned	No condition	Adjacent oracle	Mixture
2	fixed per condition	0.0625	0.1687	0.5621	0.2885	0.3119
2	fixed per condition	0.1875	0.4635	0.6671	0.0677	0.1739
2	fixed per condition	0.3125	0.5170	0.7480	0.3494	0.4125
2	fixed per condition	0.4375	0.3785	0.7775	0.5992	0.5613
2	fixed per condition	0.5625	0.3784	0.7790	0.5991	0.5613
2	fixed per condition	0.6875	0.5169	0.7499	0.3492	0.4125
2	fixed per condition	0.8125	0.4634	0.6855	0.0679	0.1741
2	fixed per condition	0.9375	0.1687	0.5752	0.2886	0.3121
3	fixed per condition	0.0625	0.1816	0.5632	0.2885	0.3341
3	fixed per condition	0.1875	0.4140	0.5397	0.0677	0.2213
3	fixed per condition	0.3125	0.4473	0.5564	0.3494	0.2114
3	fixed per condition	0.4375	0.3163	0.5536	0.1354	0.1081
3	fixed per condition	0.5625	0.3171	0.5500	0.1643	0.1297
3	fixed per condition	0.6875	0.4489	0.5307	0.3492	0.2300
3	fixed per condition	0.8125	0.4150	0.5234	0.0679	0.2286

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Table 8: Absolute secondary diagnostics for the conditioned model, averaged over Conditions and seeds. "Train" and "held" denote Training and Held-out coordinates.

n	$s/\sqrt{n}$	mean absolute error train	mean absolute error held	width absolute error train	width absolute error held	exact Conditional MMD train	exact Conditional MMD held
8	0.35	0.5797	0.2795	0.3714	0.3621	0.0184	0.0046
8	0.50	0.5964	0.3027	0.1428	0.1308	0.0193	0.0046
8	0.75	0.6319	0.3554	0.1332	0.2003	0.0208	0.0055
12	0.35	0.8565	0.3984	0.5225	0.5346	0.0204	0.0045
12	0.50	0.8969	0.4578	0.1463	0.1804	0.0220	0.0048
12	0.75	0.9641	0.5583	0.2829	0.3762	0.0244	0.0061
16	0.35	1.1360	0.5214	0.6114	0.6335	0.0206	0.0041
16	0.50	1.1909	0.6016	0.1426	0.2335	0.0228	0.0047
16	0.75	1.2908	0.7502	0.4486	0.5670	0.0259	0.0063
19	0.35	1.3490	0.6177	0.6631	0.6899	0.0204	0.0040
19	0.50	1.4049	0.7026	0.1503	0.2701	0.0228	0.0045
19	0.75	1.5071	0.8503	0.5448	0.6708	0.0263	0.0064

Table 9: Condition-count and data-allocation results at $n = 19, s = 0.50$.

K	regime	train/condition	val./condition	mean	worst	params	shared upd.	reference upd.
2	fixed per condition	10,000	2,000	0.3819	0.5170	190	2,000	4,000
3	fixed per condition	10,000	2,000	0.3402	0.4489	190	2,000	6,000
5	fixed per condition	10,000	2,000	0.1673	0.2917	190	2,000	10,000
9	fixed per condition	10,000	2,000	0.1595	0.2947	190	2,000	18,000
2	fixed total	9,000	1,800	0.3819	0.5166	190	2,000	4,000
3	fixed total	6,000	1,200	0.3399	0.4488	190	2,000	6,000
5	fixed total	3,600	720	0.1645	0.2932	190	2,000	10,000
9	fixed total	2,000	400	0.1622	0.2936	190	2,000	18,000

K	Regime	c	Conditioned	No condition	Adjacent oracle	Mixture
3	fixed per condition	0.9375	0.1813	0.5842	0.2886	0.3345
5	fixed per condition	0.0625	0.2917	0.7159	0.2885	0.3260
5	fixed per condition	0.1875	0.1247	0.5690	0.0677	0.1002
5	fixed per condition	0.3125	0.1599	0.3987	0.1638	0.0992
5	fixed per condition	0.4375	0.0925	0.2424	0.1354	0.0921
5	fixed per condition	0.5625	0.0927	0.2797	0.1643	0.1236
5	fixed per condition	0.6875	0.1601	0.4561	0.1747	0.1586
5	fixed per condition	0.8125	0.1248	0.6116	0.0679	0.0961
5	fixed per condition	0.9375	0.2916	0.7527	0.2886	0.3191
9	fixed per condition	0.0625	0.2947	0.7820	0.1439	0.2146
9	fixed per condition	0.1875	0.1120	0.6182	0.1495	0.0472
9	fixed per condition	0.3125	0.1475	0.4024	0.1621	0.0684
9	fixed per condition	0.4375	0.0835	0.1596	0.1354	0.0621
9	fixed per condition	0.5625	0.0836	0.2007	0.1507	0.0725
9	fixed per condition	0.6875	0.1478	0.4499	0.1538	0.1058
9	fixed per condition	0.8125	0.1123	0.6518	0.1417	0.1452

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Table 10: Condition-count paired profile-TVD effects with 95% intervals. Positive control, adjacent-reference-oracle, and mixture effects favor conditioning; a positive reference gap means the independent circuits have lower Training profile TVD.

K	Regime	no condition − conditioned	condi-	conditioned − reference	adjacent-reference oracle − conditioned	mixture − conditioned
2	fixed per condition	+0.0465 [+0.0289, +0.0682]	+0.0250 [+0.0221, +0.0274]	-0.0557 [-0.0697, -0.0417]	-0.0169 [-0.0309, -0.0029]	
3	fixed per condition	+0.1754 [+0.1712, +0.1797]	+0.1140 [+0.1083, +0.1194]	-0.1263 [-0.1386, -0.1117]	-0.1155 [-0.1278, -0.1008]	
5	fixed per condition	+0.2668 [+0.2587, +0.2737]	+0.0878 [+0.0860, +0.0897]	+0.0016 [-0.0012, +0.0043]	-0.0029 [-0.0053, -0.0006]	
9	fixed per condition	+0.3354 [+0.3331, +0.3379]	+0.0464 [+0.0428, +0.0497]	-0.0077 [-0.0113, -0.0028]	-0.0464 [-0.0502, -0.0412]	
2	fixed total	+0.0524 [+0.0339, +0.0724]	+0.0251 [+0.0226, +0.0273]	-0.0557 [-0.0693, -0.0424]	-0.0170 [-0.0305, -0.0036]	
3	fixed total	+0.1756 [+0.1710, +0.1799]	+0.1140 [+0.1087, +0.1192]	-0.1261 [-0.1378, -0.1121]	-0.1152 [-0.1270, -0.1013]	
5	fixed total	+0.2693 [+0.2620, +0.2739]	+0.0859 [+0.0823, +0.0892]	+0.0043 [+0.0001, +0.0091]	-0.0001 [-0.0040, +0.0042]	
9	fixed total	+0.3333 [+0.3307, +0.3358]	+0.0482 [+0.0455, +0.0510]	-0.0104 [-0.0132, -0.0076]	-0.0491 [-0.0521, -0.0463]	

K	Regime	c	Conditioned	No condition	Adjacent oracle	Mixture
9	fixed per condition	0.9375	0.2946	0.8084	0.1774	0.1890
2	fixed total	0.0625	0.1682	0.5652	0.2885	0.3119
2	fixed total	0.1875	0.4616	0.6572	0.0677	0.1739
2	fixed total	0.3125	0.5166	0.7354	0.3494	0.4125
2	fixed total	0.4375	0.3815	0.7669	0.5992	0.5613
2	fixed total	0.5625	0.3814	0.7618	0.5991	0.5613
2	fixed total	0.6875	0.5163	0.7272	0.3492	0.4125
2	fixed total	0.8125	0.4614	0.6647	0.0679	0.1741
2	fixed total	0.9375	0.1682	0.5744	0.2886	0.3121
3	fixed total	0.0625	0.1819	0.5630	0.2885	0.3341
3	fixed total	0.1875	0.4130	0.5417	0.0677	0.2213
3	fixed total	0.3125	0.4462	0.5575	0.3494	0.2114
3	fixed total	0.4375	0.3159	0.5515	0.1354	0.1081
3	fixed total	0.5625	0.3173	0.5480	0.1643	0.1297
3	fixed total	0.6875	0.4488	0.5325	0.3492	0.2300
3	fixed total	0.8125	0.4152	0.5262	0.0679	0.2286
3	fixed total	0.9375	0.1811	0.5862	0.2886	0.3345
5	fixed total	0.0625	0.2932	0.7198	0.2885	0.3260
5	fixed total	0.1875	0.1214	0.5725	0.0677	0.1002
5	fixed total	0.3125	0.1552	0.3974	0.1638	0.0992
5	fixed total	0.4375	0.0876	0.2355	0.1354	0.0921
5	fixed total	0.5625	0.0876	0.2752	0.1643	0.1236
5	fixed total	0.6875	0.1562	0.4579	0.1747	0.1586
5	fixed total	0.8125	0.1222	0.6127	0.0679	0.0961

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K	Regime	c	Conditioned	No condition	Adjacent oracle	Mixture
5	fixed total	0.9375	0.2926	0.7549	0.2886	0.3191
9	fixed total	0.0625	0.2936	0.7815	0.1439	0.2146
9	fixed total	0.1875	0.1150	0.6173	0.1495	0.0472
9	fixed total	0.3125	0.1519	0.4010	0.1621	0.0684
9	fixed total	0.4375	0.0872	0.1583	0.1354	0.0621
9	fixed total	0.5625	0.0878	0.2006	0.1507	0.0725
9	fixed total	0.6875	0.1532	0.4498	0.1538	0.1058
9	fixed total	0.8125	0.1157	0.6518	0.1417	0.1452
9	fixed total	0.9375	0.2933	0.8087	0.1774	0.1890

G Exhaustive paired intervals

The following machine-generated table lists every diagnostic comparison, not only the profile-TVD effects selected for the main text. The replication unit is always the matched optimizer seed; conditions are averaged before resampling.

Table 12: Paired intervals for $n = 8$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0243 [-0.0445, -0.0054]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	-0.0005 [-0.0006, -0.0004]	unsupported
heldout neighbor oracle minus conditioned mean absolute error	+0.1103 [+0.0770, +0.1392]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0292 [-0.0468, -0.0108]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.0962 [-0.1592, -0.0276]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0328 [-0.0540, -0.0126]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0006 [-0.0008, -0.0005]	unsupported
heldout mixture minus conditioned mean absolute error	-0.0030 [-0.0251, +0.0171]	mixed
heldout mixture minus conditioned profile tvd	-0.0248 [-0.0418, -0.0064]	unsupported
heldout mixture minus conditioned width absolute error	-0.0194 [-0.0820, +0.0489]	mixed
heldout no condition minus conditioned bitstring tvd	+0.2735 [+0.2572, +0.2906]	supported
heldout no condition minus conditioned conditional mmd	+0.0406 [+0.0405, +0.0407]	supported
heldout no condition minus conditioned mean absolute error	+1.1205 [+1.0980, +1.1410]	supported
heldout no condition minus conditioned profile tvd	+0.2840 [+0.2695, +0.2990]	supported
heldout no condition minus conditioned width absolute error	+0.7802 [+0.6850, +0.8592]	supported
training conditioned minus independent reference bitstring tvd	+0.0662 [+0.0474, +0.0856]	supported
training conditioned minus independent reference conditional mmd	+0.0074 [+0.0073, +0.0076]	supported
training conditioned minus independent reference mean absolute error	+0.1592 [+0.1460, +0.1735]	supported
training conditioned minus independent reference profile tvd	+0.0923 [+0.0766, +0.1065]	supported
training conditioned minus independent reference width absolute error	+0.0722 [+0.0179, +0.1220]	supported
training no condition minus conditioned bitstring tvd	+0.2388 [+0.2232, +0.2554]	supported
training no condition minus conditioned conditional mmd	+0.0525 [+0.0520, +0.0530]	supported
training no condition minus conditioned mean absolute error	+1.1017 [+1.0862, +1.1164]	supported
training no condition minus conditioned profile tvd	+0.2402 [+0.2278, +0.2536]	supported
training no condition minus conditioned width absolute error	+0.7978 [+0.7088, +0.8691]	supported

Table 13: Paired intervals for $n = 8$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0380 [-0.0491, -0.0277]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	-0.0005 [-0.0006, -0.0003]	unsupported
heldout neighbor oracle minus conditioned mean absolute error	+0.0431 [+0.0287, +0.0570]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0280 [-0.0336, -0.0225]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.0796 [-0.0909, -0.0692]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0362 [-0.0473, -0.0263]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0003 [-0.0005, -0.0002]	unsupported
heldout mixture minus conditioned mean absolute error	-0.0377 [-0.0500, -0.0250]	unsupported
heldout mixture minus conditioned profile tvd	-0.0363 [-0.0432, -0.0292]	unsupported
heldout mixture minus conditioned width absolute error	-0.0028 [-0.0168, +0.0109]	mixed
heldout no condition minus conditioned bitstring tvd	+0.2375 [+0.2166, +0.2641]	supported
heldout no condition minus conditioned conditional mmd	+0.0407 [+0.0405, +0.0409]	supported
heldout no condition minus conditioned mean absolute error	+1.0973 [+1.0828, +1.1116]	supported
heldout no condition minus conditioned profile tvd	+0.2698 [+0.2542, +0.2965]	supported
heldout no condition minus conditioned width absolute error	+0.8385 [+0.7480, +0.8905]	supported
training conditioned minus independent reference bitstring tvd	+0.0411 [+0.0310, +0.0525]	supported
training conditioned minus independent reference conditional mmd	+0.0070 [+0.0069, +0.0071]	supported
training conditioned minus independent reference mean absolute error	+0.1843 [+0.1766, +0.1915]	supported
training conditioned minus independent reference profile tvd	+0.0788 [+0.0718, +0.0857]	supported
training conditioned minus independent reference width absolute error	+0.0366 [+0.0221, +0.0507]	supported
training no condition minus conditioned bitstring tvd	+0.2055 [+0.1855, +0.2306]	supported
training no condition minus conditioned conditional mmd	+0.0514 [+0.0509, +0.0522]	supported
training no condition minus conditioned mean absolute error	+1.0859 [+1.0750, +1.0977]	supported
training no condition minus conditioned profile tvd	+0.2170 [+0.2015, +0.2418]	supported
training no condition minus conditioned width absolute error	+0.8941 [+0.7986, +0.9503]	supported

Table 14: Paired intervals for $n = 8$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0099 [-0.0165, -0.0032]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	+0.0015 [+0.0014, +0.0016]	supported
heldout neighbor oracle minus conditioned mean absolute error	+0.1133 [+0.1009, +0.1227]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0276 [-0.0339, -0.0214]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.0902 [-0.1037, -0.0750]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0596 [-0.0653, -0.0544]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0005 [-0.0006, -0.0005]	unsupported
heldout mixture minus conditioned mean absolute error	-0.1080 [-0.1201, -0.0993]	unsupported
heldout mixture minus conditioned profile tvd	-0.0525 [-0.0584, -0.0461]	unsupported
heldout mixture minus conditioned width absolute error	-0.1657 [-0.1743, -0.1574]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.1557 [+0.1278, +0.1969]	supported
heldout no condition minus conditioned conditional mmd	+0.0398 [+0.0395, +0.0404]	supported
heldout no condition minus conditioned mean absolute error	+1.0446 [+1.0327, +1.0538]	supported
heldout no condition minus conditioned profile tvd	+0.1900 [+0.1717, +0.2167]	supported
heldout no condition minus conditioned width absolute error	+0.5997 [+0.5068, +0.6518]	supported
training conditioned minus independent reference bitstring tvd	+0.0503 [+0.0447, +0.0560]	supported
training conditioned minus independent reference conditional mmd	+0.0067 [+0.0066, +0.0068]	supported
training conditioned minus independent reference mean absolute error	+0.2277 [+0.2216, +0.2352]	supported
training conditioned minus independent reference profile tvd	+0.0688 [+0.0635, +0.0742]	supported
training conditioned minus independent reference width absolute error	+0.0995 [+0.0901, +0.1085]	supported
training no condition minus conditioned bitstring tvd	+0.1288 [+0.1018, +0.1699]	supported
training no condition minus conditioned conditional mmd	+0.0495 [+0.0490, +0.0506]	supported
training no condition minus conditioned mean absolute error	+1.0511 [+1.0412, +1.0592]	supported
training no condition minus conditioned profile tvd	+0.1460 [+0.1293, +0.1714]	supported
training no condition minus conditioned width absolute error	+0.7865 [+0.6961, +0.8358]	supported

Table 15: Paired intervals for $n = 12$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0371 [-0.0466, -0.0271]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	-0.0004 [-0.0005, -0.0003]	unsupported
heldout neighbor oracle minus conditioned mean absolute error	+0.0992 [+0.0886, +0.1100]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0221 [-0.0255, -0.0190]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.1215 [-0.1299, -0.1129]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0153 [-0.0247, -0.0051]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0000 [-0.0001, +0.0001]	mixed
heldout mixture minus conditioned mean absolute error	+0.0297 [+0.0162, +0.0431]	supported
heldout mixture minus conditioned profile tvd	-0.0016 [-0.0052, +0.0019]	mixed
heldout mixture minus conditioned width absolute error	+0.0261 [+0.0165, +0.0359]	supported
heldout no condition minus conditioned bitstring tvd	+0.3019 [+0.2894, +0.3164]	supported
heldout no condition minus conditioned conditional mmd	+0.0410 [+0.0409, +0.0411]	supported
heldout no condition minus conditioned mean absolute error	+1.7016 [+1.6905, +1.7132]	supported
heldout no condition minus conditioned profile tvd	+0.2892 [+0.2798, +0.2998]	supported
heldout no condition minus conditioned width absolute error	+0.5399 [+0.3070, +0.7984]	supported
training conditioned minus independent reference bitstring tvd	+0.0509 [+0.0423, +0.0587]	supported
training conditioned minus independent reference conditional mmd	+0.0074 [+0.0073, +0.0075]	supported
training conditioned minus independent reference mean absolute error	+0.2118 [+0.2023, +0.2218]	supported
training conditioned minus independent reference profile tvd	+0.0831 [+0.0791, +0.0872]	supported
training conditioned minus independent reference width absolute error	+0.0752 [+0.0675, +0.0830]	supported
training no condition minus conditioned bitstring tvd	+0.2578 [+0.2494, +0.2684]	supported
training no condition minus conditioned conditional mmd	+0.0552 [+0.0541, +0.0562]	supported
training no condition minus conditioned mean absolute error	+1.6805 [+1.6708, +1.6909]	supported
training no condition minus conditioned profile tvd	+0.2419 [+0.2352, +0.2505]	supported
training no condition minus conditioned width absolute error	+0.5786 [+0.3458, +0.8372]	supported

Table 16: Paired intervals for $n = 12$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0356 [-0.0470, -0.0249]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	-0.0001 [-0.0002, -0.0000]	unsupported
heldout neighbor oracle minus conditioned mean absolute error	+0.0568 [+0.0496, +0.0653]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0180 [-0.0213, -0.0146]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.1159 [-0.1234, -0.1074]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0215 [-0.0328, -0.0107]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0001 [+0.0000, +0.0002]	supported
heldout mixture minus conditioned mean absolute error	-0.0563 [-0.0635, -0.0469]	unsupported
heldout mixture minus conditioned profile tvd	-0.0251 [-0.0283, -0.0216]	unsupported
heldout mixture minus conditioned width absolute error	+0.0116 [+0.0043, +0.0190]	supported
heldout no condition minus conditioned bitstring tvd	+0.3024 [+0.2842, +0.3197]	supported
heldout no condition minus conditioned conditional mmd	+0.0419 [+0.0415, +0.0424]	supported
heldout no condition minus conditioned mean absolute error	+1.6422 [+1.6358, +1.6508]	supported
heldout no condition minus conditioned profile tvd	+0.2793 [+0.2663, +0.2923]	supported
heldout no condition minus conditioned width absolute error	+0.8925 [+0.5774, +1.1911]	supported
training conditioned minus independent reference bitstring tvd	+0.0312 [+0.0220, +0.0405]	supported
training conditioned minus independent reference conditional mmd	+0.0071 [+0.0070, +0.0072]	supported
training conditioned minus independent reference mean absolute error	+0.2773 [+0.2705, +0.2830]	supported
training conditioned minus independent reference profile tvd	+0.0877 [+0.0839, +0.0911]	supported
training conditioned minus independent reference width absolute error	+0.0343 [+0.0266, +0.0403]	supported
training no condition minus conditioned bitstring tvd	+0.2594 [+0.2388, +0.2782]	supported
training no condition minus conditioned conditional mmd	+0.0542 [+0.0524, +0.0560]	supported
training no condition minus conditioned mean absolute error	+1.6407 [+1.6322, +1.6518]	supported
training no condition minus conditioned profile tvd	+0.2291 [+0.2213, +0.2391]	supported
training no condition minus conditioned width absolute error	+1.0030 [+0.6903, +1.2993]	supported

Table 17: Paired intervals for $n = 12$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	+0.0156 [+0.0083, +0.0233]	supported
heldout neighbor oracle minus conditioned conditional mmd	+0.0018 [+0.0014, +0.0023]	supported
heldout neighbor oracle minus conditioned mean absolute error	+0.1243 [+0.0983, +0.1512]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0129 [-0.0151, -0.0107]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.0644 [-0.0878, -0.0402]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0476 [-0.0501, -0.0453]	unsupported
heldout mixture minus conditioned conditional mmd	-0.0001 [-0.0001, -0.0000]	unsupported
heldout mixture minus conditioned mean absolute error	-0.1791 [-0.1851, -0.1731]	unsupported
heldout mixture minus conditioned profile tvd	-0.0548 [-0.0567, -0.0526]	unsupported
heldout mixture minus conditioned width absolute error	-0.2365 [-0.2563, -0.2149]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.1953 [+0.1689, +0.2248]	supported
heldout no condition minus conditioned conditional mmd	+0.0408 [+0.0398, +0.0420]	supported
heldout no condition minus conditioned mean absolute error	+1.5417 [+1.5374, +1.5463]	supported
heldout no condition minus conditioned profile tvd	+0.2019 [+0.1903, +0.2138]	supported
heldout no condition minus conditioned width absolute error	+0.8031 [+0.5245, +1.0192]	supported
training conditioned minus independent reference bitstring tvd	+0.0288 [+0.0256, +0.0321]	supported
training conditioned minus independent reference conditional mmd	+0.0066 [+0.0066, +0.0066]	supported
training conditioned minus independent reference mean absolute error	+0.3499 [+0.3473, +0.3527]	supported
training conditioned minus independent reference profile tvd	+0.0582 [+0.0554, +0.0613]	supported
training conditioned minus independent reference width absolute error	+0.0797 [+0.0647, +0.0938]	supported
training no condition minus conditioned bitstring tvd	+0.1687 [+0.1409, +0.1999]	supported
training no condition minus conditioned conditional mmd	+0.0500 [+0.0480, +0.0524]	supported
training no condition minus conditioned mean absolute error	+1.5664 [+1.5605, +1.5728]	supported
training no condition minus conditioned profile tvd	+0.1608 [+0.1515, +0.1722]	supported
training no condition minus conditioned width absolute error	+1.0492 [+0.7734, +1.2647]	supported

Table 18: Paired intervals for $n = 16$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0429 [-0.0538, -0.0319]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	-0.0001 [-0.0001, +0.0000]	mixed
heldout neighbor oracle minus conditioned mean absolute error	+0.1446 [+0.1381, +0.1499]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0229 [-0.0279, -0.0174]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.1356 [-0.1557, -0.1146]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0060 [-0.0166, +0.0048]	mixed
heldout mixture minus conditioned conditional mmd	+0.0004 [+0.0003, +0.0005]	supported
heldout mixture minus conditioned mean absolute error	+0.0464 [+0.0369, +0.0546]	supported
heldout mixture minus conditioned profile tvd	+0.0117 [+0.0067, +0.0172]	supported
heldout mixture minus conditioned width absolute error	+0.0987 [+0.0783, +0.1208]	supported
heldout no condition minus conditioned bitstring tvd	+0.3087 [+0.2927, +0.3256]	supported
heldout no condition minus conditioned conditional mmd	+0.0399 [+0.0397, +0.0400]	supported
heldout no condition minus conditioned mean absolute error	+2.2786 [+2.2711, +2.2848]	supported
heldout no condition minus conditioned profile tvd	+0.3139 [+0.3059, +0.3231]	supported
heldout no condition minus conditioned width absolute error	+0.6858 [+0.2739, +1.1408]	supported
training conditioned minus independent reference bitstring tvd	+0.0464 [+0.0370, +0.0554]	supported
training conditioned minus independent reference conditional mmd	+0.0068 [+0.0067, +0.0069]	supported
training conditioned minus independent reference mean absolute error	+0.2799 [+0.2737, +0.2868]	supported
training conditioned minus independent reference profile tvd	+0.0835 [+0.0785, +0.0878]	supported
training conditioned minus independent reference width absolute error	+0.0675 [+0.0499, +0.0838]	supported
training no condition minus conditioned bitstring tvd	+0.2515 [+0.2406, +0.2632]	supported
training no condition minus conditioned conditional mmd	+0.0538 [+0.0525, +0.0550]	supported
training no condition minus conditioned mean absolute error	+2.2557 [+2.2469, +2.2646]	supported
training no condition minus conditioned profile tvd	+0.2472 [+0.2421, +0.2519]	supported
training no condition minus conditioned width absolute error	+0.7318 [+0.3236, +1.1828]	supported

Table 19: Paired intervals for $n = 16$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0418 [-0.0510, -0.0318]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	+0.0001 [+0.0001, +0.0002]	supported
heldout neighbor oracle minus conditioned mean absolute error	+0.0876 [+0.0800, +0.0953]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0040 [-0.0071, +0.0001]	mixed
heldout neighbor oracle minus conditioned width absolute error	-0.1386 [-0.1487, -0.1256]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0193 [-0.0282, -0.0093]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0004 [+0.0003, +0.0005]	supported
heldout mixture minus conditioned mean absolute error	-0.0697 [-0.0805, -0.0566]	unsupported
heldout mixture minus conditioned profile tvd	-0.0099 [-0.0133, -0.0057]	unsupported
heldout mixture minus conditioned width absolute error	+0.0298 [+0.0163, +0.0472]	supported
heldout no condition minus conditioned bitstring tvd	+0.3069 [+0.2923, +0.3265]	supported
heldout no condition minus conditioned conditional mmd	+0.0408 [+0.0403, +0.0413]	supported
heldout no condition minus conditioned mean absolute error	+2.1984 [+2.1900, +2.2077]	supported
heldout no condition minus conditioned profile tvd	+0.3138 [+0.3019, +0.3270]	supported
heldout no condition minus conditioned width absolute error	+0.9599 [+0.4548, +1.4967]	supported
training conditioned minus independent reference bitstring tvd	+0.0352 [+0.0264, +0.0432]	supported
training conditioned minus independent reference conditional mmd	+0.0067 [+0.0066, +0.0068]	supported
training conditioned minus independent reference mean absolute error	+0.3687 [+0.3605, +0.3757]	supported
training conditioned minus independent reference profile tvd	+0.0933 [+0.0895, +0.0964]	supported
training conditioned minus independent reference width absolute error	+0.0278 [+0.0091, +0.0434]	supported
training no condition minus conditioned bitstring tvd	+0.2575 [+0.2402, +0.2765]	supported
training no condition minus conditioned conditional mmd	+0.0534 [+0.0513, +0.0554]	supported
training no condition minus conditioned mean absolute error	+2.2020 [+2.1924, +2.2116]	supported
training no condition minus conditioned profile tvd	+0.2440 [+0.2354, +0.2537]	supported
training no condition minus conditioned width absolute error	+1.1301 [+0.6214, +1.6714]	supported

Table 20: Paired intervals for $n = 16$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	+0.0250 [+0.0190, +0.0313]	supported
heldout neighbor oracle minus conditioned conditional mmd	+0.0029 [+0.0026, +0.0032]	supported
heldout neighbor oracle minus conditioned mean absolute error	+0.2461 [+0.2180, +0.2729]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0192 [-0.0205, -0.0178]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.1705 [-0.1893, -0.1523]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0559 [-0.0594, -0.0518]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0001 [+0.0001, +0.0002]	supported
heldout mixture minus conditioned mean absolute error	-0.2429 [-0.2518, -0.2340]	unsupported
heldout mixture minus conditioned profile tvd	-0.0681 [-0.0696, -0.0666]	unsupported
heldout mixture minus conditioned width absolute error	-0.3812 [-0.4014, -0.3606]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.2406 [+0.2034, +0.2781]	supported
heldout no condition minus conditioned conditional mmd	+0.0405 [+0.0394, +0.0417]	supported
heldout no condition minus conditioned mean absolute error	+2.0498 [+2.0416, +2.0579]	supported
heldout no condition minus conditioned profile tvd	+0.1981 [+0.1780, +0.2176]	supported
heldout no condition minus conditioned width absolute error	+0.8736 [+0.4476, +1.2787]	supported
training conditioned minus independent reference bitstring tvd	+0.0258 [+0.0219, +0.0293]	supported
training conditioned minus independent reference conditional mmd	+0.0064 [+0.0064, +0.0064]	supported
training conditioned minus independent reference mean absolute error	+0.4633 [+0.4530, +0.4731]	supported
training conditioned minus independent reference profile tvd	+0.0523 [+0.0495, +0.0553]	supported
training conditioned minus independent reference width absolute error	+0.1339 [+0.1182, +0.1497]	supported
training no condition minus conditioned bitstring tvd	+0.2100 [+0.1710, +0.2486]	supported
training no condition minus conditioned conditional mmd	+0.0499 [+0.0473, +0.0526]	supported
training no condition minus conditioned mean absolute error	+2.1053 [+2.0940, +2.1171]	supported
training no condition minus conditioned profile tvd	+0.1667 [+0.1570, +0.1770]	supported
training no condition minus conditioned width absolute error	+1.1659 [+0.7386, +1.5723]	supported

Table 21: Paired intervals for $n = 19$ and $s = 0.35\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0610 [-0.0776, -0.0458]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	-0.0001 [-0.0003, +0.0001]	mixed
heldout neighbor oracle minus conditioned mean absolute error	+0.1758 [+0.1493, +0.2119]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0217 [-0.0240, -0.0192]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.1371 [-0.1597, -0.1162]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0175 [-0.0342, -0.0021]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0003 [+0.0001, +0.0005]	supported
heldout mixture minus conditioned mean absolute error	+0.0533 [+0.0228, +0.0942]	supported
heldout mixture minus conditioned profile tvd	+0.0188 [+0.0164, +0.0213]	supported
heldout mixture minus conditioned width absolute error	+0.1641 [+0.1409, +0.1852]	supported
heldout no condition minus conditioned bitstring tvd	+0.2943 [+0.2769, +0.3101]	supported
heldout no condition minus conditioned conditional mmd	+0.0389 [+0.0386, +0.0391]	supported
heldout no condition minus conditioned mean absolute error	+2.7073 [+2.6799, +2.7452]	supported
heldout no condition minus conditioned profile tvd	+0.3356 [+0.3282, +0.3468]	supported
heldout no condition minus conditioned width absolute error	+0.4266 [+0.1216, +0.9637]	supported
training conditioned minus independent reference bitstring tvd	+0.0620 [+0.0488, +0.0763]	supported
training conditioned minus independent reference conditional mmd	+0.0065 [+0.0064, +0.0067]	supported
training conditioned minus independent reference mean absolute error	+0.3355 [+0.3104, +0.3554]	supported
training conditioned minus independent reference profile tvd	+0.0870 [+0.0847, +0.0893]	supported
training conditioned minus independent reference width absolute error	+0.0574 [+0.0395, +0.0769]	supported
training no condition minus conditioned bitstring tvd	+0.2394 [+0.2256, +0.2523]	supported
training no condition minus conditioned conditional mmd	+0.0533 [+0.0519, +0.0542]	supported
training no condition minus conditioned mean absolute error	+2.6746 [+2.6561, +2.6983]	supported
training no condition minus conditioned profile tvd	+0.2620 [+0.2594, +0.2645]	supported
training no condition minus conditioned width absolute error	+0.4747 [+0.1704, +1.0099]	supported

Table 22: Paired intervals for $n = 19$ and $s = 0.50\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	-0.0606 [-0.0749, -0.0461]	unsupported
heldout neighbor oracle minus conditioned conditional mmd	+0.0002 [+0.0002, +0.0003]	supported
heldout neighbor oracle minus conditioned mean absolute error	+0.1145 [+0.1000, +0.1286]	supported
heldout neighbor oracle minus conditioned profile tvd	+0.0016 [-0.0012, +0.0043]	mixed
heldout neighbor oracle minus conditioned width absolute error	-0.1451 [-0.1555, -0.1358]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0300 [-0.0443, -0.0155]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0005 [+0.0005, +0.0006]	supported
heldout mixture minus conditioned mean absolute error	-0.0696 [-0.0804, -0.0585]	unsupported
heldout mixture minus conditioned profile tvd	-0.0029 [-0.0053, -0.0006]	unsupported
heldout mixture minus conditioned width absolute error	+0.0521 [+0.0416, +0.0622]	supported
heldout no condition minus conditioned bitstring tvd	+0.3037 [+0.2885, +0.3183]	supported
heldout no condition minus conditioned conditional mmd	+0.0400 [+0.0394, +0.0404]	supported
heldout no condition minus conditioned mean absolute error	+2.6224 [+2.6098, +2.6350]	supported
heldout no condition minus conditioned profile tvd	+0.3360 [+0.3223, +0.3507]	supported
heldout no condition minus conditioned width absolute error	+0.8220 [+0.2164, +1.5105]	supported
training conditioned minus independent reference bitstring tvd	+0.0428 [+0.0296, +0.0560]	supported
training conditioned minus independent reference conditional mmd	+0.0064 [+0.0063, +0.0064]	supported
training conditioned minus independent reference mean absolute error	+0.4279 [+0.4206, +0.4354]	supported
training conditioned minus independent reference profile tvd	+0.0878 [+0.0860, +0.0897]	supported
training conditioned minus independent reference width absolute error	+0.0319 [+0.0135, +0.0514]	supported
training no condition minus conditioned bitstring tvd	+0.2559 [+0.2370, +0.2723]	supported
training no condition minus conditioned conditional mmd	+0.0530 [+0.0507, +0.0549]	supported
training no condition minus conditioned mean absolute error	+2.6314 [+2.6155, +2.6474]	supported
training no condition minus conditioned profile tvd	+0.2668 [+0.2587, +0.2737]	supported
training no condition minus conditioned width absolute error	+1.0212 [+0.4158, +1.7108]	supported

Table 23: Paired intervals for $n = 19$ and $s = 0.75\sqrt{n}$. Positive effects favor the conditioned model under the named subtraction.

Comparison	Effect [95% interval]	Status
heldout neighbor oracle minus conditioned bitstring tvd	+0.0074 [-0.0032, +0.0165]	mixed
heldout neighbor oracle minus conditioned conditional mmd	+0.0020 [+0.0016, +0.0023]	supported
heldout neighbor oracle minus conditioned mean absolute error	+0.2478 [+0.2051, +0.2986]	supported
heldout neighbor oracle minus conditioned profile tvd	-0.0111 [-0.0165, -0.0049]	unsupported
heldout neighbor oracle minus conditioned width absolute error	-0.1981 [-0.2308, -0.1643]	unsupported
heldout mixture minus conditioned bitstring tvd	-0.0656 [-0.0768, -0.0563]	unsupported
heldout mixture minus conditioned conditional mmd	+0.0002 [-0.0002, +0.0004]	mixed
heldout mixture minus conditioned mean absolute error	-0.2483 [-0.2763, -0.2077]	unsupported
heldout mixture minus conditioned profile tvd	-0.0619 [-0.0670, -0.0560]	unsupported
heldout mixture minus conditioned width absolute error	-0.4633 [-0.4961, -0.4286]	unsupported
heldout no condition minus conditioned bitstring tvd	+0.2272 [+0.1839, +0.2702]	supported
heldout no condition minus conditioned conditional mmd	+0.0401 [+0.0384, +0.0418]	supported
heldout no condition minus conditioned mean absolute error	+2.4747 [+2.4455, +2.5170]	supported
heldout no condition minus conditioned profile tvd	+0.2261 [+0.2088, +0.2418]	supported
heldout no condition minus conditioned width absolute error	+0.8519 [+0.1776, +1.5129]	supported
training conditioned minus independent reference bitstring tvd	+0.0385 [+0.0306, +0.0474]	supported
training conditioned minus independent reference conditional mmd	+0.0063 [+0.0062, +0.0066]	supported
training conditioned minus independent reference mean absolute error	+0.5383 [+0.5122, +0.5573]	supported
training conditioned minus independent reference profile tvd	+0.0603 [+0.0544, +0.0659]	supported
training conditioned minus independent reference width absolute error	+0.1626 [+0.1321, +0.1922]	supported
training no condition minus conditioned bitstring tvd	+0.1940 [+0.1476, +0.2405]	supported
training no condition minus conditioned conditional mmd	+0.0503 [+0.0465, +0.0542]	supported
training no condition minus conditioned mean absolute error	+2.5272 [+2.5060, +2.5506]	supported
training no condition minus conditioned profile tvd	+0.1807 [+0.1648, +0.1999]	supported
training no condition minus conditioned width absolute error	+1.1021 [+0.3924, +1.7942]	supported

H Archived Gaussian ablations

The topology, encoding, and ancilla studies predate the frozen factorial and are preserved as bounded ablations. Their metrics are regenerated from the archived JSON/NPZ pairs and are not pooled with the new campaign. Complete connectivity uses 190 phases at $n = 19$; the lattice and heavy-hex banks use 48 and 39. The symmetric-reupload comparison uses three matched seeds. The ancilla ladder uses $m \in \{0, 1, 2, 3, 5, 7\}$ and compares each augmented model with a visible-only matched-parameter control.

Permutation twirling replaces a raw sample x by $\pi(x)$ for a uniformly random qubit permutation π. For an exchangeable target, this preserves the Hamming-weight profile and makes the output uniform within each observed shell in expectation. It is therefore a legitimate target-symmetry-aware post-processing baseline. It is not a symmetry learned by the circuit, it uses known target structure, and it is excluded from raw-circuit claims.

I Negative two-coordinate feasibility study

An archived eight-qubit pilot attempted to condition jointly on Gaussian location and width. Its frozen design required single-condition representability before any shared-model or interpolation interpretation. At seed 0, the four single-condition bitstring TVDs were 0.3883, 0.3177, 0.2220, and 0.0945. The two narrow-width cells failed the declared capacity screen; their width errors were 0.4659 and 0.1058. The study stopped, so it supplies no evidence for two-coordinate conditional generation. We include it to mark the boundary of the present one-coordinate claim and to prevent selective reporting of only successful Gaussian variants.

J Campaign integrity and reproduction

The Gaussian campaign is isolated under `results/gaussian-foundations-v1`. The immutable manifest digest binds the run plan; a source-tree digest binds the numerical core used by the resource preflight and every result, while a companion dependency-closure snapshot binds the encoder, ansatz, simulator, target, artifact, environment, and lockfile sources. Dataset archives record role, derived seed, condition grid, master-pool hash, and split hash. A result is complete only when its envelope, checkpoint, and probability archive all exist and their digests agree. Reservations without a complete optimizer state are never silently restarted.

From the repository root, the principal commands are:

```
uv sync --frozen
uv run python scripts/prepare_gaussian_campaign.py
uv run python scripts/run_gaussian_campaign.py preflight
uv run python scripts/freeze_gaussian_execution_source.py
uv run python scripts/run_gaussian_campaign.py run \
  --shard-count 5 --shard-index 0
uv run python scripts/summarize_gaussian_campaign.py
uv run python scripts/verify_gaussian_legacy.py
uv run python scripts/finalize_gaussian_claims.py
uv run python scripts/generate_gaussian_tables.py
uv run python scripts/plot_gaussian_campaign.py
uv run pytest -q
uv run ruff check src scripts tests
uv run python scripts/build_gaussian_arxiv_archive.py
uv run python scripts/audit_gaussian_release.py
uv run python scripts/prepare_curie_weiss_campaign.py
uv run python scripts/run_curie_weiss_campaign.py preflight
uv run python scripts/run_curie_weiss_campaign.py run \
  --shard-count 5 --shard-index 0
uv run python scripts/finalize_curie_weiss_campaign.py
uv run python scripts/plot_curie_weiss_campaign.py
```

Shard indices 1–4 are executed analogously. Re-running a completed shard verifies and skips immutable outputs. Report, claim, table, and figure generators refuse partial campaign state.